INDUCTION HOB

BASIC	:	C61R1AAMST
MODEL	:	CTI613EHST
MODEL CODE	:	CTI613EHST/XEF

SERVICE Manual

INDUCTION HOB



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2. Product Specification
3. Disassembly and Reassembly
4. Troubleshooting
5. Exploded Views and Part List
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Refer to the service manual in the GSPN(see rear cover) for the more information.

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1. General Safety Precautions

1. Information contained in this manual is intended for use by a qualified service technician. The technician is required to be familiar with proper and safe procedures to be followed when repairing appliance. All tests and repairs are to be performed using proper tools and measuring devices. All component replacements should be made **using only factory approved replacement parts**.

2. Electrical shock and injury can result if service or repairs are attempted by an unqualified individual. **Improper disassembly, assembly or adjustments can create hazardous conditions**.

3. Even for a skilled technician, a risk of injury or electrical shock exists while performing service or repairs. Electrical injury can be serious or fatal. **Extreme caution must taken when performing voltage checks on individual components of an appliance**.

NOTE

Except as necessary to perform a particular step in servicing a product, the electrical ***power supply should always be disconnected when servicing a product.***

4. To avoid possible injury, the appliance must be properly grounded. Never plug in or direct wire an appliance unless it is **properly grounded. See the installation instructions that originally accompanied the product for proper grounding procedures**.

5.The ventilation gap between the worktop and front of the unit underneath it must not be covered.

6. Before connecting, check that the nominal voltage of the appliance, that is, the voltage stated on the rating plate, corresponds to the available supply voltage. The rating plate is located on the lower casing of the hob.

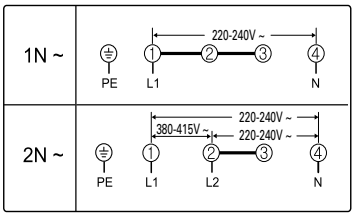
7. The heating element voltage is AC230V~. The appliance also works perfectly on networks with AC220V~ or AC240V~.

8. The hob is to be connected to the mains using a device that allows the appliance to be disconnected from the mains at all poles with a contact opening width of at least 3 mm, eg. automatic line protecting cut - out, earth leakage trips or fuse.

9. A type H05RN - F cable or one of a higher grade must be used as the mains connecting cable.

	Rated current of appliance (A)	Nominal cross-sectional area (mm^2)
1N~	> 25 and ≤ 32	≥2.5
2N~	> 10 and ≤ 16	≥1.5

10. The connection must be carried out as shown in the diagram. The connecting links should be fitted according to the appropriate connection diagram.



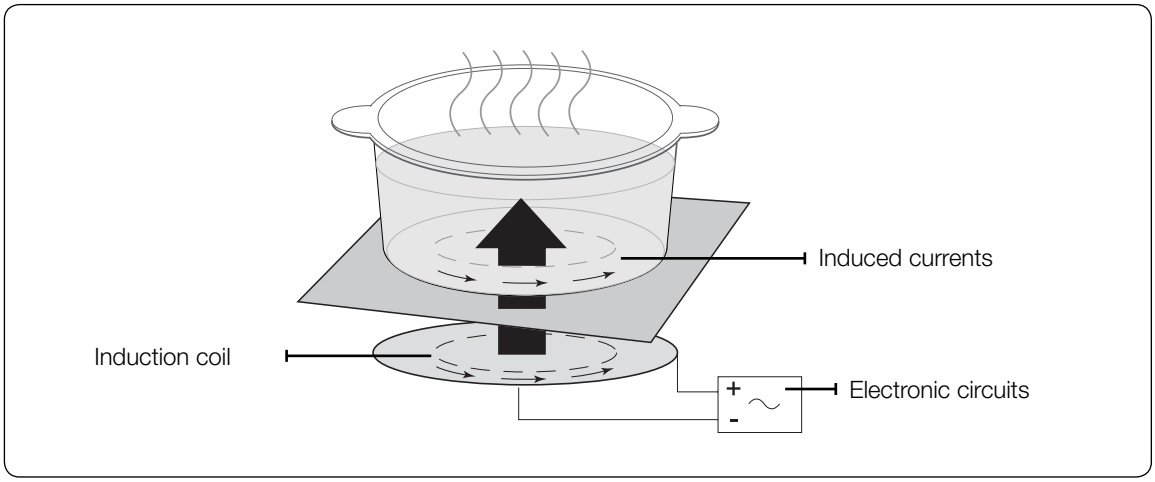
< IN ~ >
L: Brown
N: Blue
PE: Green

⚠ In case of 1N~, connect the hob to more than 35A. When connecting the hob to less than 35A, make sure that circuit breaker can operate.

2. Specifications

2-1 Induction Heating

- **The Principle of Induction Heating**: When you place your cookware on a cooking zone and you turn it on, the electronic circuits in your induction hob produce “induced currents” in the bottom of the cookware which instantly raise cookware’s temperature.



2-2 Features

	Basic model	New model
Image		
Model Name	C61R1AAMST/XEU	CTI613EHST/XEF
Product Type	Radiant Hob	Induction Hob
Glass	Ceramic Glass	Ceramic Glass
Control Method	Touch Sensor	Touch Sensor
Residual Heat Indicator	Yes	Yes
Safety Shutoff	Yes	Yes
Pan Detection	-	Yes
Install Type	Built-In	Built-In
Design	4-side STSS	4-side STSS
Electric Features		
Burner Power		
Front-Left	1,800W	1,400W/Boost 2,000W
Rear-Left	1,200W	2,200W/Boost 3,200W
Front-Right	1,200W	1,800W/Boost 2,600W
Rear-Right	1,800W	1,800W/Boost 2,600W

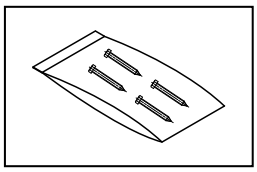
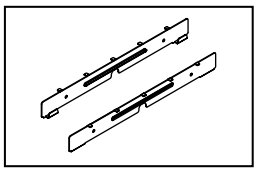
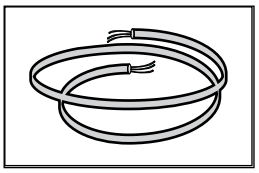
2. Specifications

	Basic model	New model
Burner Size		
Front-Left	180mm	160mm
Rear-Left	145mm	210mm
Front-Right	145mm	180mm
Rear-Right	180mm	180mm
Control Features		
Child Safety Lock	Yes	Yes
Countdown Timer	-	Yes
Safety Timer	-	Yes
Sound On/Off	-	-
Power Level	1~9 level	1~15 level
Boost	-	Yes
Power Pre-Selection	-	Yes

2-3 Table of Specification

	Basic model	New model
Electrical Connection		
Rated voltage	220-240V~ 50/60Hz	220-240V~ 50/60Hz
Mains-connection	2N : 380-415V~ 50/60Hz	2N : 380-415V~ 50/60Hz
	1N : 220-240V~ 50/60Hz	1N : 220-240V~ 50/60Hz
Output Power	MAX 5,500 - 6,500W	MAX 7,200W
Outside Dimensions(WxLxH)	575 x 505 x 55mm	575 x 505 x 62mm
Net Weight	7.5Kg	12.0Kg
Gross Weight	10.0Kg	14.5Kg
Export Zone	UK	France

2-4 Accessory

		
Screw (DG97-00010A)	Installation Bracket (DG61-00005B)	Power Cable (DG96-00061B)

3. Disassembly and Reassembly

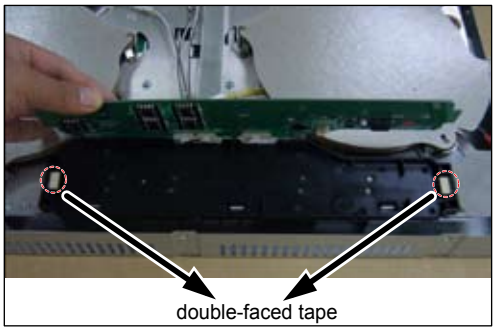
3-1 Replacement of the Assy Top Plate

Attention

The Ceramic Glass may break if you use force especially on the edge.

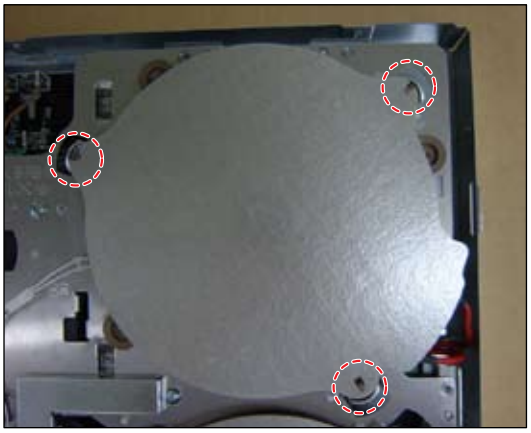
Parts	Explanation Photo	Explanation
Assy Top Plate (Ceramic Glass)		1. Disconnect power. 2. Remove 4 - side direction 12 Screws securing the Assy Top Plate to the Assy Case Burner. 3. Raise the Assy Top Plate. 4. After replacement of the Assy Top Plate, Screw the Assy Top Plate to the Assy Case Burner.

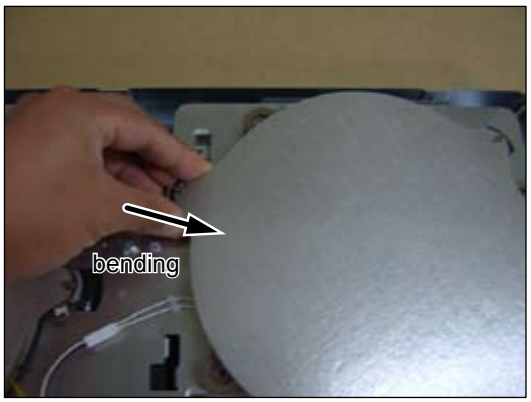
3-2 Replacement of the Assy Control Panel

Parts	Explanation Photo	Explanation
Assy Control Panel		1. Lift up the Assy Touch PCB. (Assy Touch PCB is fixed by double-faced tape) 2. Disconnect all sub-wire. 3. Replace the Assy Touch PCB or Control-Panel.

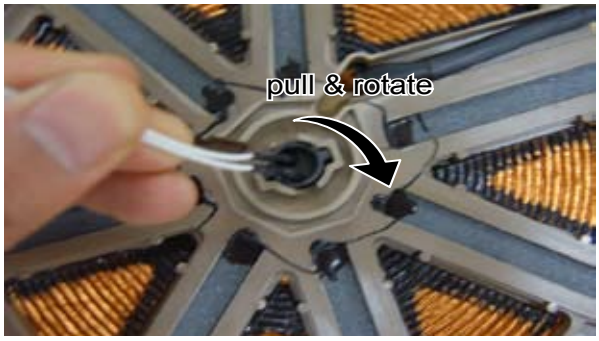
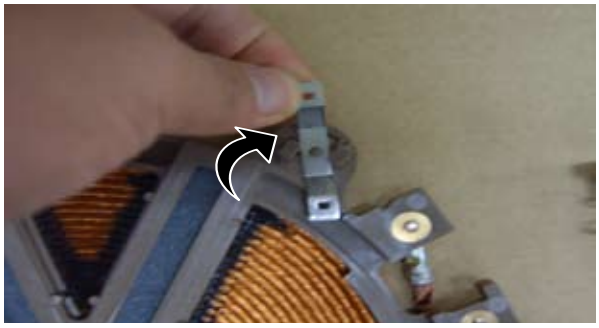
3-3 Replacement of the Assy-Working Coil

<table>
<tr><th colspan="3">Explanation Photo</th></tr>
<tr><td colspan="3">

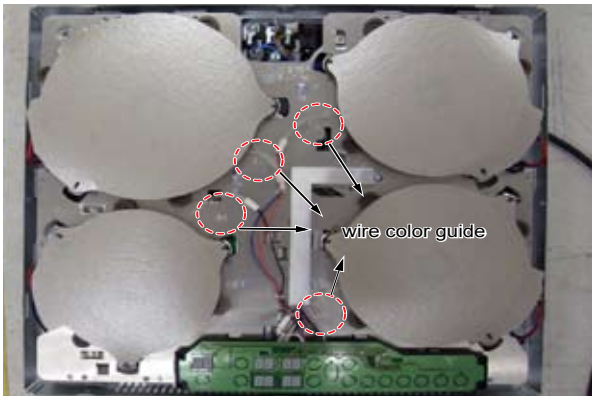
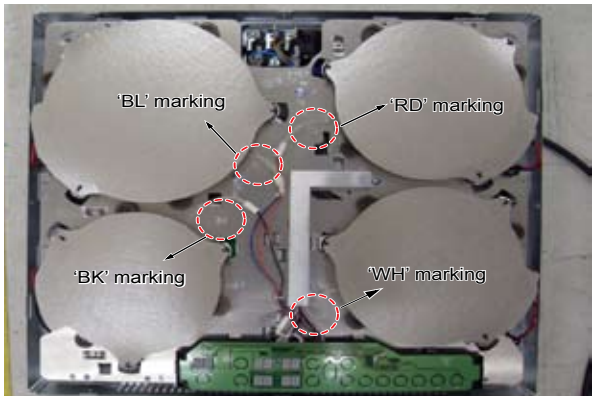

</td></tr>
<tr><th>Parts</th><th>Explanation Photo</th><th>Explanation</th></tr>
<tr><td rowspan="2">Assy-Working Coil</td><td>  </td><td rowspan="2">1. Bend the Cover-Coil a little.

2. Lift up the Cover-Coil from the hook. (3-point)</td></tr>
<tr><td> 
</td></tr>
</table>

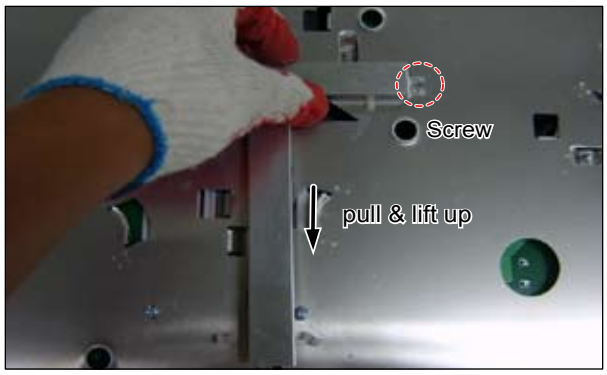
3-3 Replacement of the Assy-Working Coil

Parts	Explanation Photo	Explanation
Assy-Working Coil		3. Remove the Adiabatic-Coil by using some tool. (Do not remove by hand. Adiabatic can be easily broken.) 4. Disconnect all lead wires from the Assy-Working Coil.
		5. For the replacement of Sensor-Top, pull the Sensor-Top toward bottom side. 6. Rotate the Sensor-Wire by 90degree until the Sensor-Top can be remove from the Coil-Working.
		6. Rotate the Spring ETC-Base by 90degree in order to replace.

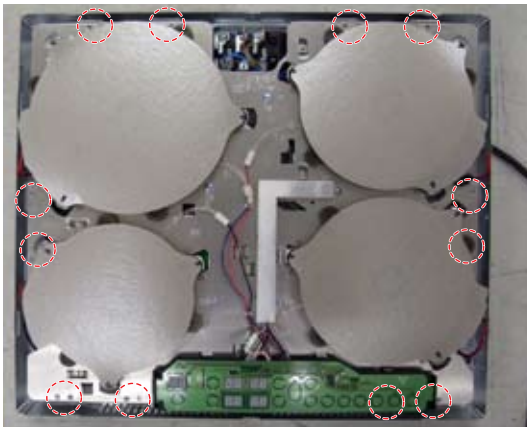
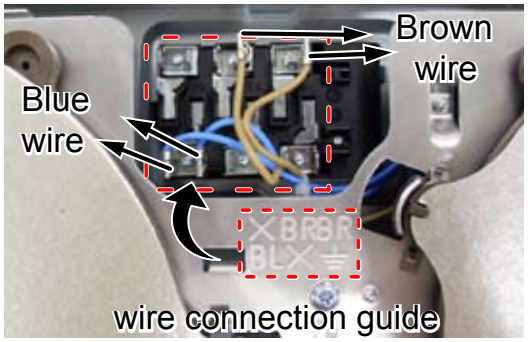
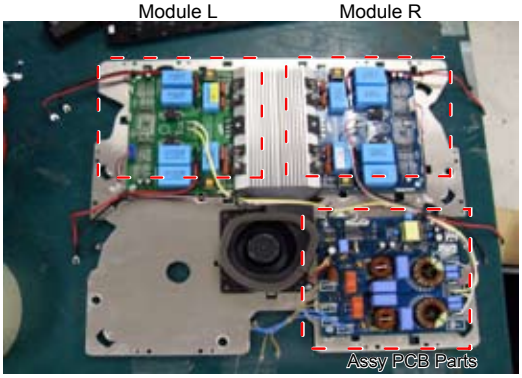
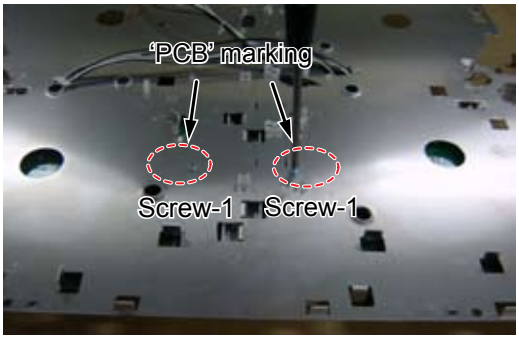
3-3 Replacement of the Assy-Working Coil

<table>
<tr><th>Parts</th><th>Explanation Photo</th><th>Explanation</th></tr>
<tr><td rowspan="3">Assy-Working Coil</td><td> 
</td><td rowspan="2">7. After replacement, connect the Sensor-Top wire according to carved wire color guide.</td></tr>
<tr><td> 
</td></tr>
<tr><td colspan="2">warning
Wire connecting position is very improtant.
Please keep attention.</td></tr>
</table>

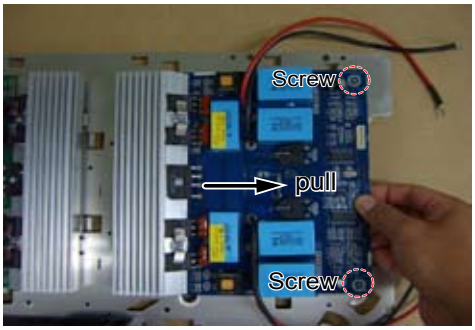
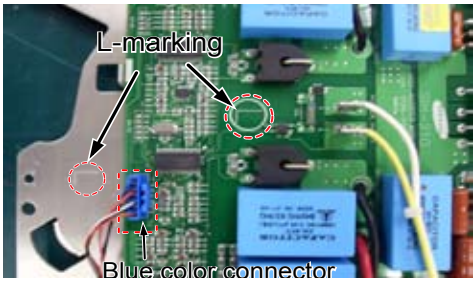
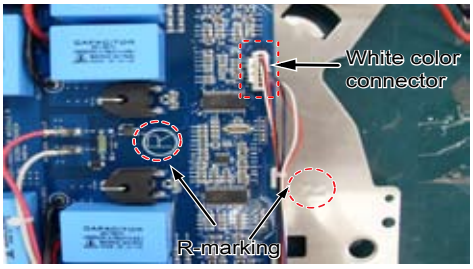
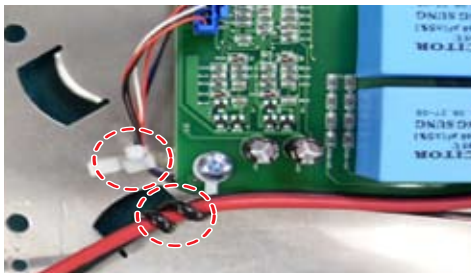
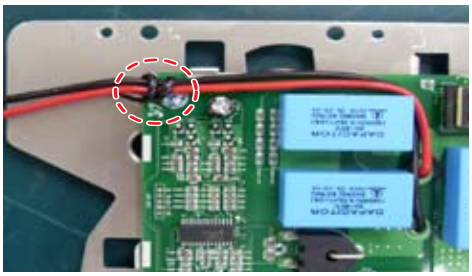
3-4 Replacement of the Bracket-Wire A

Parts	Explanation Photo	Explanation
Bracket-Wire A		1. Remove the screw. 2. Pull the Bracket-Wire A toward front side. 3. Lift up the Bracket-Wire A. 4. Replace the Bracket-Wire A or sub-wire.
	warning Wire can be damaged when the repalcement of Bracket-Wire A. Please keep attention.	

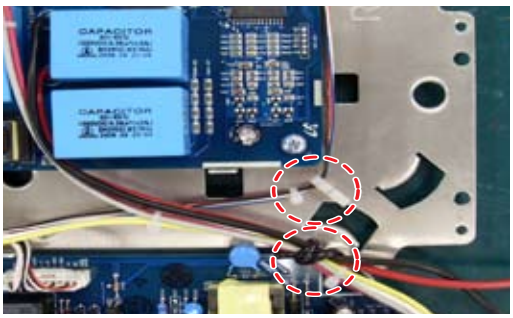
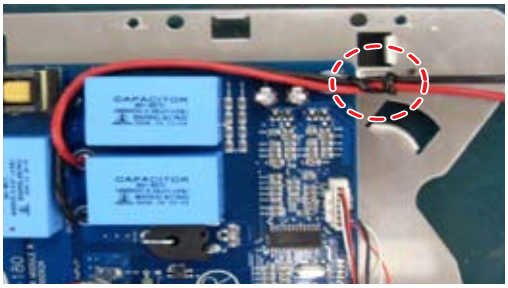
3-5 Replacement of the Assy-Induction Module

Parts	Explanation Photo	Explanation
Assy-Induction Module		1. Remove 12 screws(4-side direction) securing the Assy-Induction Module to the Assy Case Burner.
		2. Disconnect Assy Wire Harness-P from the Terminal Block. 3. After replacement, connect the wire according to carved guide.
	Warning Wire connecting position is very improtant. Please keep attention.	
	 	4. Remove the screw(upper side) securing the Assy-Inverter Module to the Case-Induction A. * screw-1 type : 6002-000630 (M3,L8) * 'PCB' marking is carved at the side of screw.

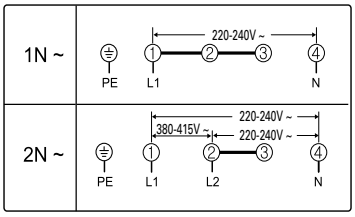
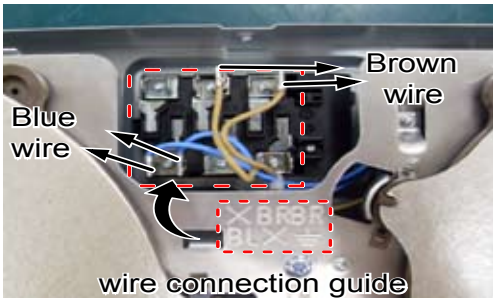
3-5 Replacement of the Assy-Induction Module

Parts	Explanation Photo	Explanation
Assy-Induction Module		5. Remove the 2 screw. 6. Pull the Assy-Inverter Module(L,R) toward left or right side. 7. Replace the Assy-Inverter Module(L,R).
		8.Arrange Assy-Inverter Module(L,R) to proper position. (Please see the picture.)
		9. Connect the sub-wire with matched color position. (Please see the picture.)
		10.After replacement, arrange all wire properly by using Holder-Wie Saddle.
		11. Especially, wire of Assy-Inverter Module(L,R) must be secured to proper position by using Cable Clamp. (See the picture)

3-5 Replacement of the Assy-Induction Module

<table>
<tr><td rowspan="4">Assy-Induction Module</td><td> 
  </td><td></td></tr>
<tr><td colspan="2">Attention
If wire of Assy-Inverter Module(L,R) is connected to wrong Assy-Working Coil, wrong cooking zone will be operated when costomer start the cook.</td></tr>
<tr><td> 
</td><td>12. Remove the 1 screw at Motor DC.(bottom side)
* screw-2 type : 6006-001170 (M4,L10,Ground screw)</td></tr>
<tr><td> 
</td><td>13. Remove the 2 screws at Motor DC.(upper side)
* ‘FM’ marking is carved at the upper side.
* screw-3 type : 6002-001321 (M5,L10)
14. Replace the Motor DC.</td></tr>
</table>

3-6 Replacement of the Assy Case Burner

Parts	Explanation Photo	Explanation
Assy Case Burner		1. Open the cover & lock of Terminal Block by using small flat type screw driver.
		2. Disconnect the power cable.
	 < IN ~ > L: Brown N: Blue PE: Green	3. After replacement of Terminal Block, connect the power cable properly.
		4. Connect the inner side of Terminal Block's wire properly.

4. Troubleshooting

4-1 Part Checking method

Parts	Photo	Good	No Good
Sensor-Top		7.65KΩ ~ 9.34KΩ (at 25˚C)	The others
Motor DC		1.65K~1.8KΩ (between No.1 pin and No.3 pin)	The others
Coil-Working		0.01~1Ω	100MΩ exceed

4-2 Failure Codes

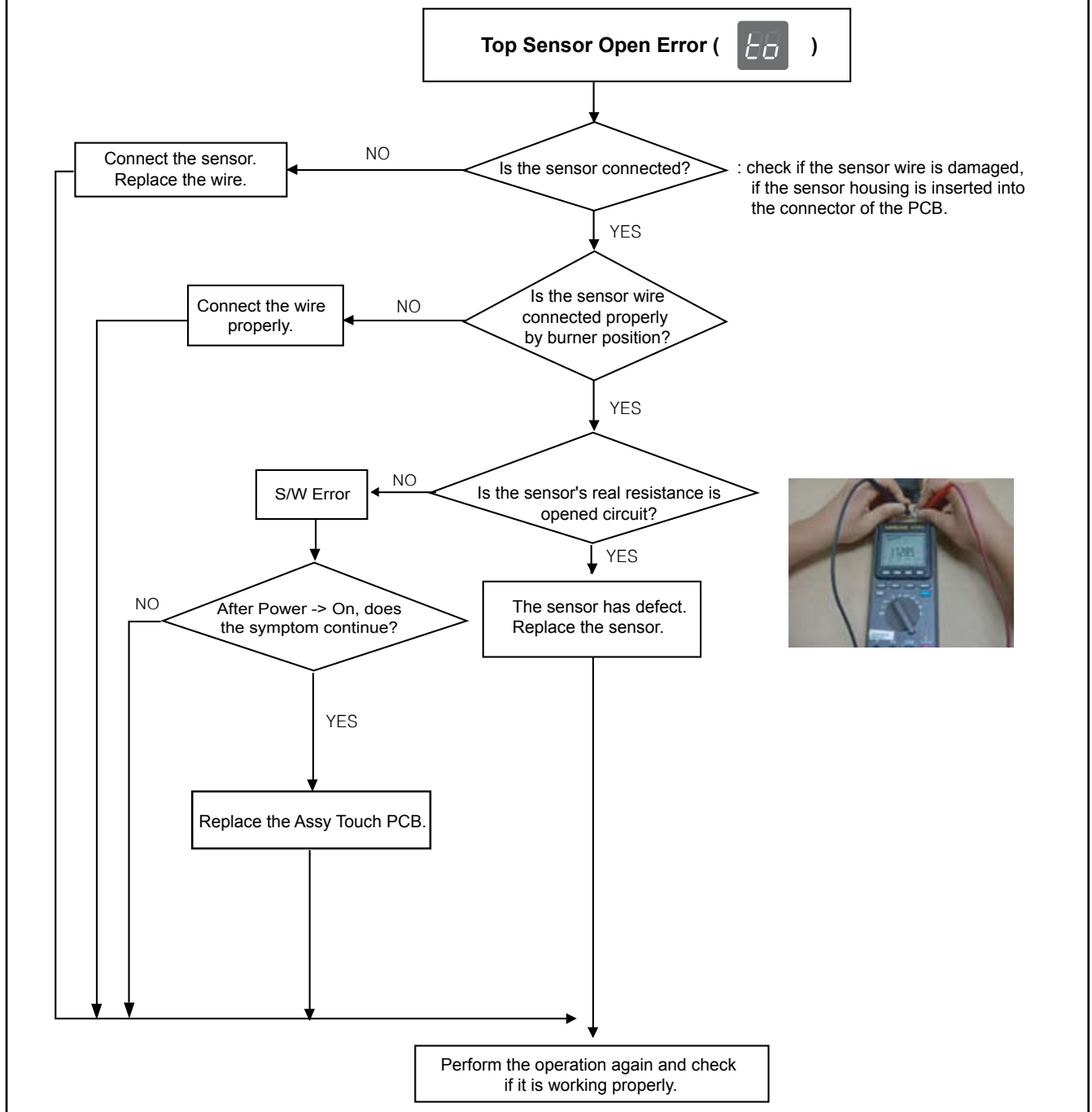
4-2-1 Temp Sensor Error

Error Code	Solution	Page
	**Top Sensor Open Error (Sensor-Top)** It occurs due to a defective sensor, misplaced wires, a defective PCB and when A/D value that MICOM senses rises over 252. Also, it may occur when the ambient temperature falls under -10˚C.	16 Page
	**Top Sensor Short Error (Sensor-Top)** It occurs due to a defective sensor, misplaced wires, a defective PCB and when A/D value that MICOM senses falls under 10.	17 Page
	**IGBT Sensor Open Error (Assy-Inverter Module)** It occurs due to a defective sensor, misplaced wires, a defective PCB and when A/D value that MICOM senses rises over 239. Also, it may occur when the ambient temperature falls under -10˚C.	18 Page
	**IGBT Sensor Short Error (Assy-Inverter Module)** It occurs due to a defective sensor, misplaced wires, a defective PCB and when A/D value that MICOM senses falls under 10.	19 Page

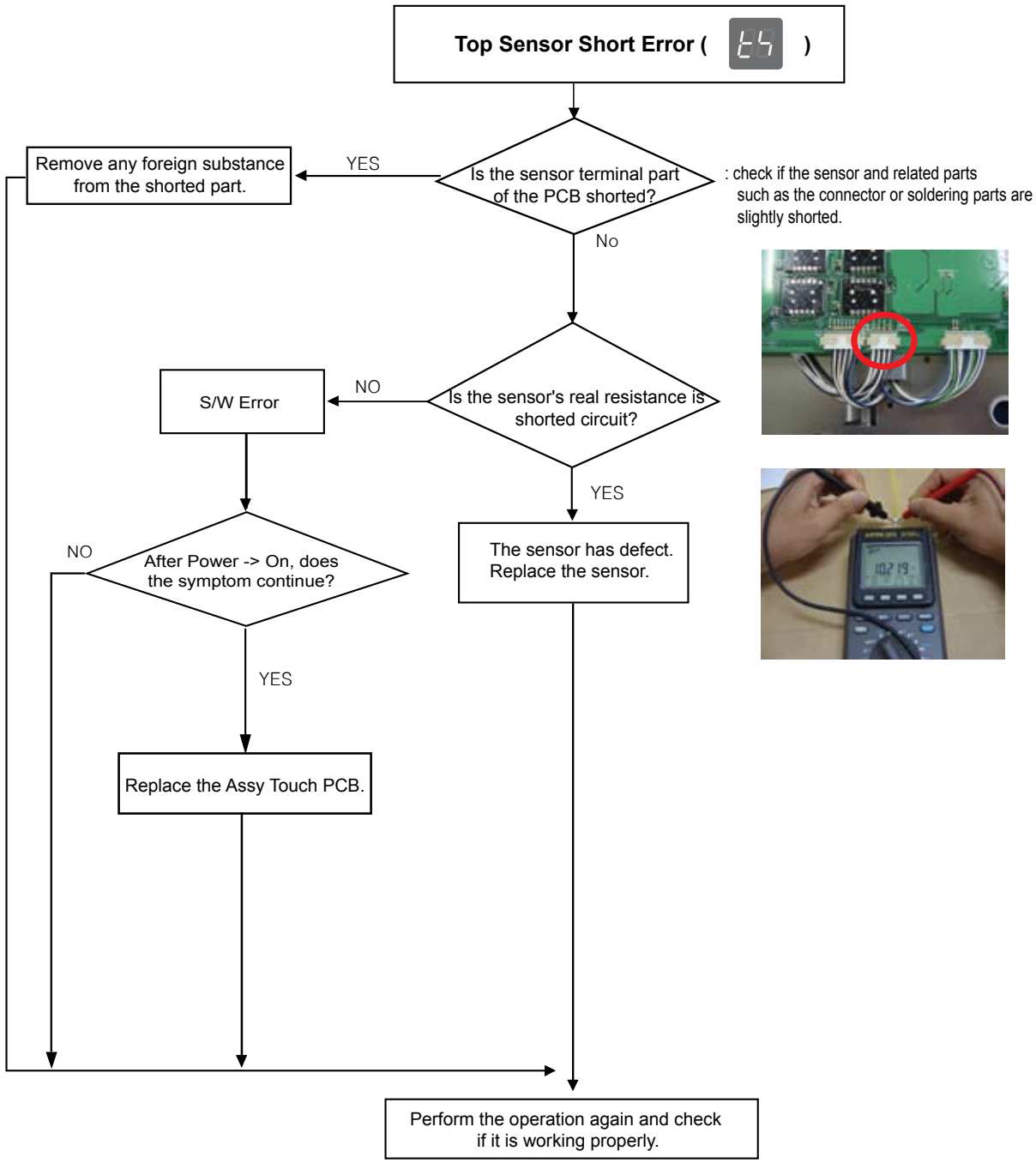
4-2-2 Safety Error

Error Code	Gerneral Function	Solution	Page
	Touch Button Short Error	It occurs when the control panel's sensor field is shorted more than 10 seconds. ex: No.1 Place a damp cloth on the control panel. No.2 Liquid boils over and lands on the control panel. No.3 More than one key is pressed more than 10 seconds. No.4 Defective Assy-Touch PCB	20 Page
	Low Voltage Error	It occurs when the DC 12V is dropped under 9V. It may occurs due to defects of PCB or wiring.	21 Page
	Over Temperature Error	It occurs when the temperature of the Top Sensor rises very highly. (Estimated temperature of ceramic glass's surface is more than 250˚C.) ex: Place a empty cookware on the burner and operate the hob.	22 Page
	Pan Detection Error	It occurs when the cookware is unsuitable or too small or no cookware has been placed on the cooking zone. If the suitable cookware is placed again, the hob will operate normally.	23 Page
	DC Motor Locking Error	It occurs when the DC Motor cannot operate due to defects of PCB, wiring or some disturbance on motor blade.	24 Page

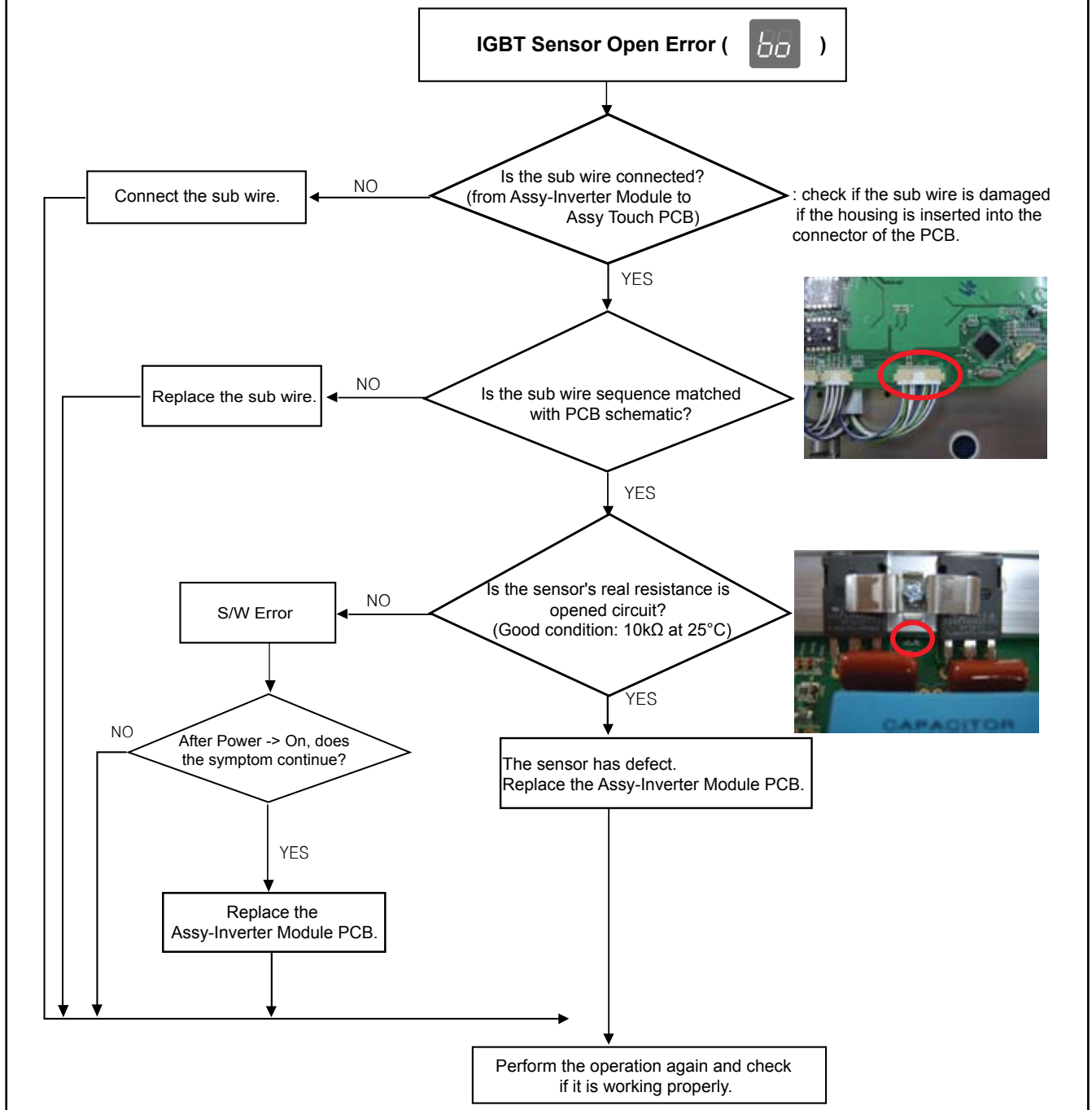
4-2-3 tO


Top Sensor Open Error (tO)
Is the sensor connected?
NO
Connect the sensor.
Replace the wire.
: check if the sensor wire is damaged,
if the sensor housing is inserted into
the connector of the PCB.
YES
Is the sensor wire
connected properly
by burner position?
NO
Connect the wire
properly.
YES
Is the sensor's real resistance is
opened circuit?
NO
S/W Error
YES
The sensor has defect.
Replace the sensor.
After Power -> On, does
the symptom continue?
NO
YES
Replace the Assy Touch PCB.
Perform the operation again and check
if it is working properly.

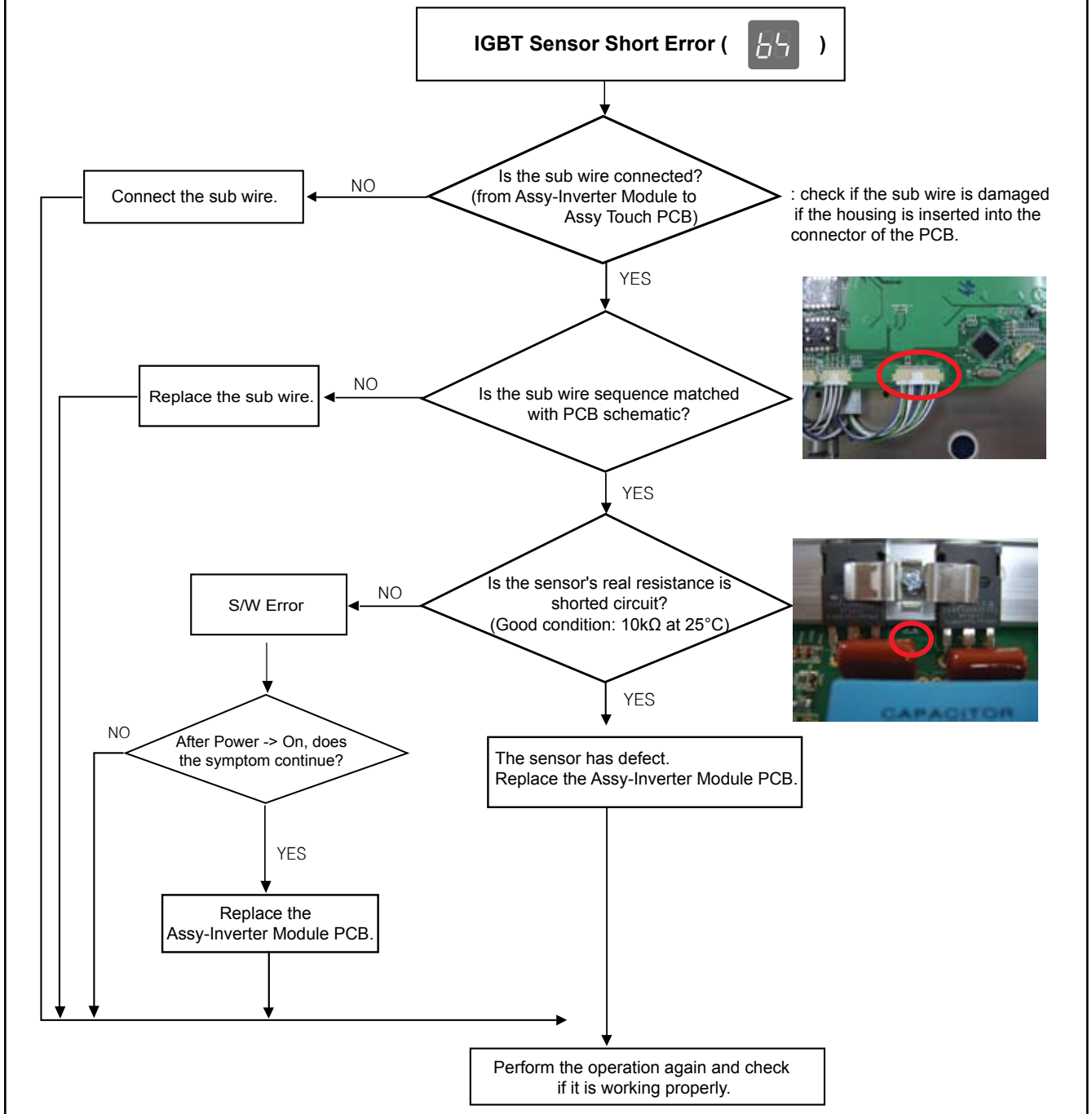
4-2-4 E4


Top Sensor Short Error (E4)
Is the sensor terminal part of the PCB shorted?
YES
Remove any foreign substance from the shorted part.
: check if the sensor and related parts such as the connector or soldering parts are slightly shorted.
No
Is the sensor's real resistance is shorted circuit?
NO
S/W Error
YES
The sensor has defect. Replace the sensor.
NO
After Power -> On, does the symptom continue?
YES
Replace the Assy Touch PCB.
Perform the operation again and check if it is working properly.

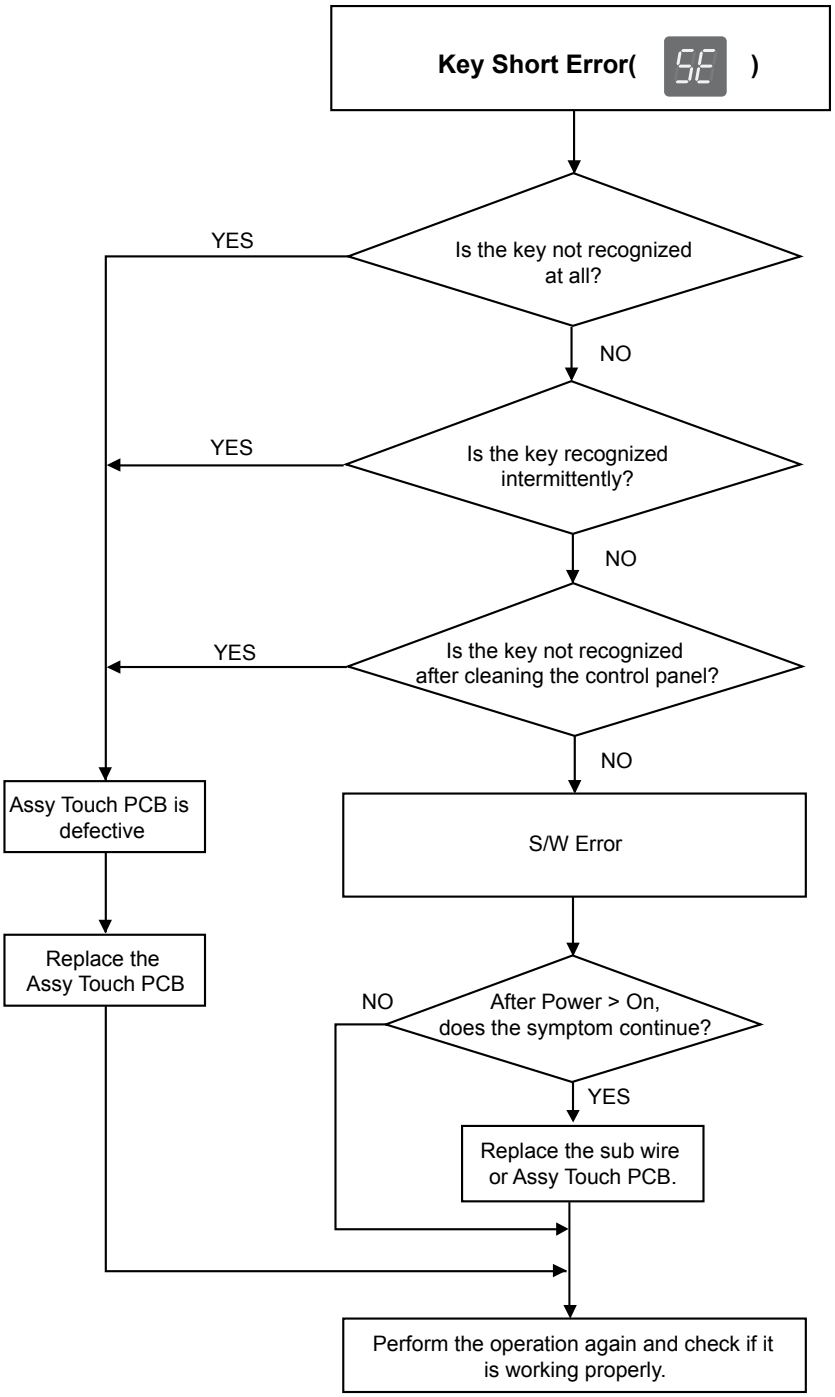
4-2-5 bb


IGBT Sensor Open Error (bb)
Is the sub wire connected?
(from Assy-Inverter Module to
Assy Touch PCB)
NO
Connect the sub wire.
: check if the sub wire is damaged
if the housing is inserted into the
connector of the PCB.
YES
Is the sub wire sequence matched
with PCB schematic?
NO
Replace the sub wire.
YES
Is the sensor's real resistance is
opened circuit?
(Good condition: 10kΩ at 25°C)
NO
S/W Error
YES
The sensor has defect.
Replace the Assy-Inverter Module PCB.
After Power -> On, does
the symptom continue?
NO
YES
Replace the
Assy-Inverter Module PCB.
Perform the operation again and check
if it is working properly.
CAPACITOR

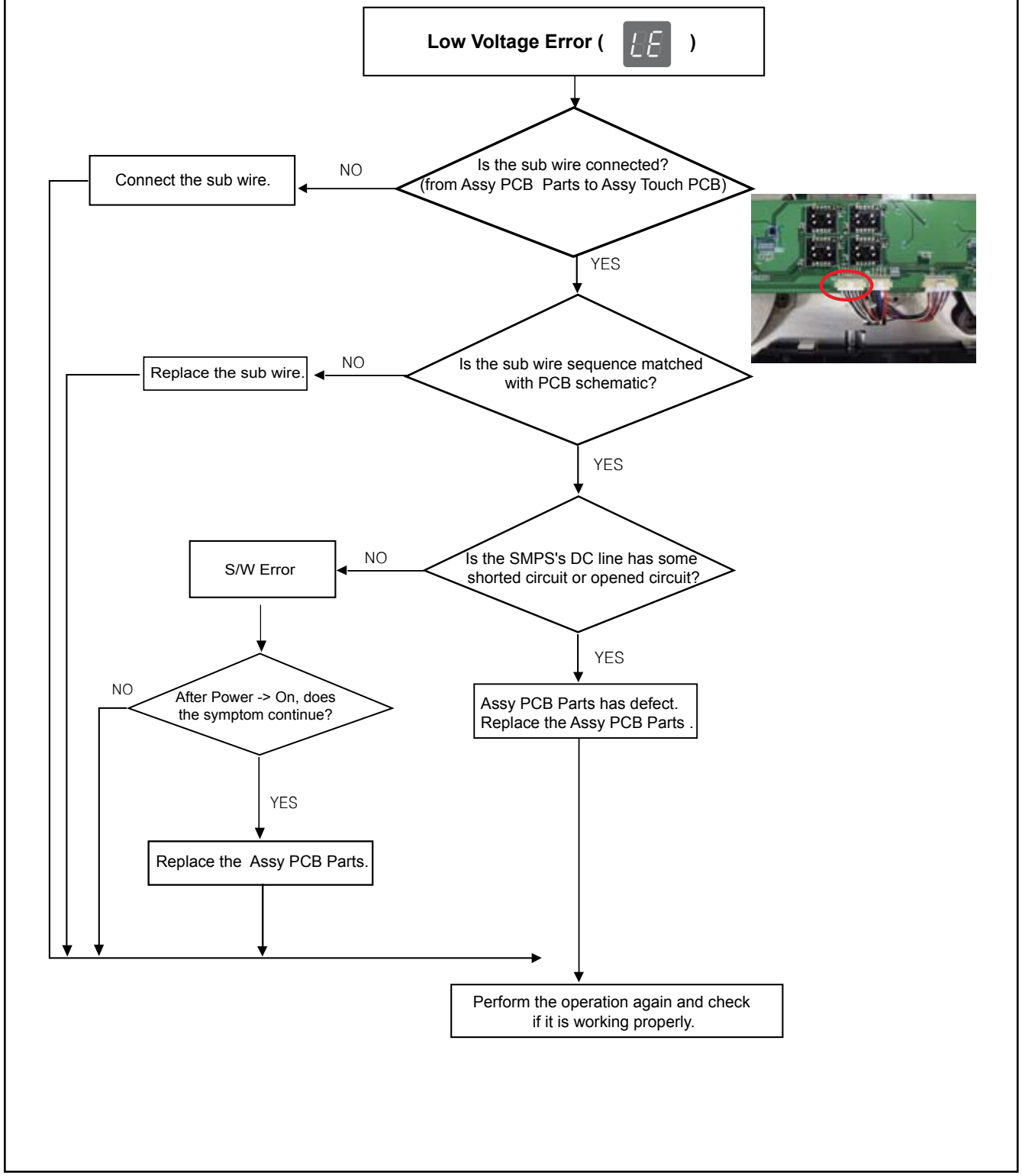
4-2-6 64


IGBT Sensor Short Error (64)
Is the sub wire connected? (from Assy-Inverter Module to Assy Touch PCB)
NO
Connect the sub wire.
: check if the sub wire is damaged if the housing is inserted into the connector of the PCB.
YES
Is the sub wire sequence matched with PCB schematic?
NO
Replace the sub wire.
YES
Is the sensor's real resistance is shorted circuit? (Good condition: 10kΩ at 25°C)
NO
S/W Error
YES
The sensor has defect. Replace the Assy-Inverter Module PCB.
After Power -> On, does the symptom continue?
NO
YES
Replace the Assy-Inverter Module PCB.
Perform the operation again and check if it is working properly.
CAPACITOR

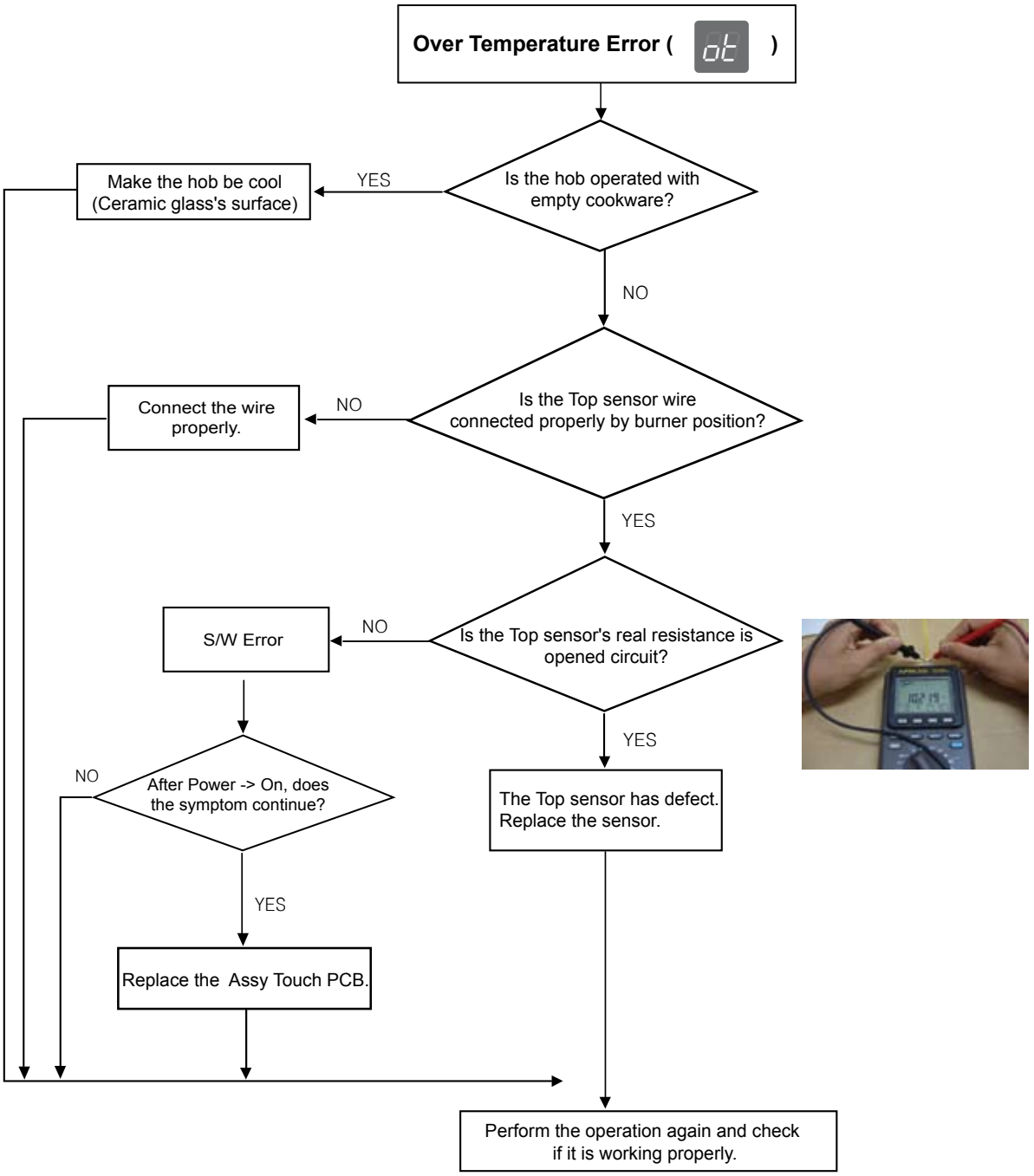
4-2-7 SE


Key Short Error(SE)
Is the key not recognized at all?
YES
NO
Is the key recognized intermittently?
YES
NO
Is the key not recognized after cleaning the control panel?
YES
NO
Assy Touch PCB is defective
S/W Error
Replace the Assy Touch PCB
After Power > On, does the symptom continue?
NO
YES
Replace the sub wire or Assy Touch PCB.
Perform the operation again and check if it is working properly.

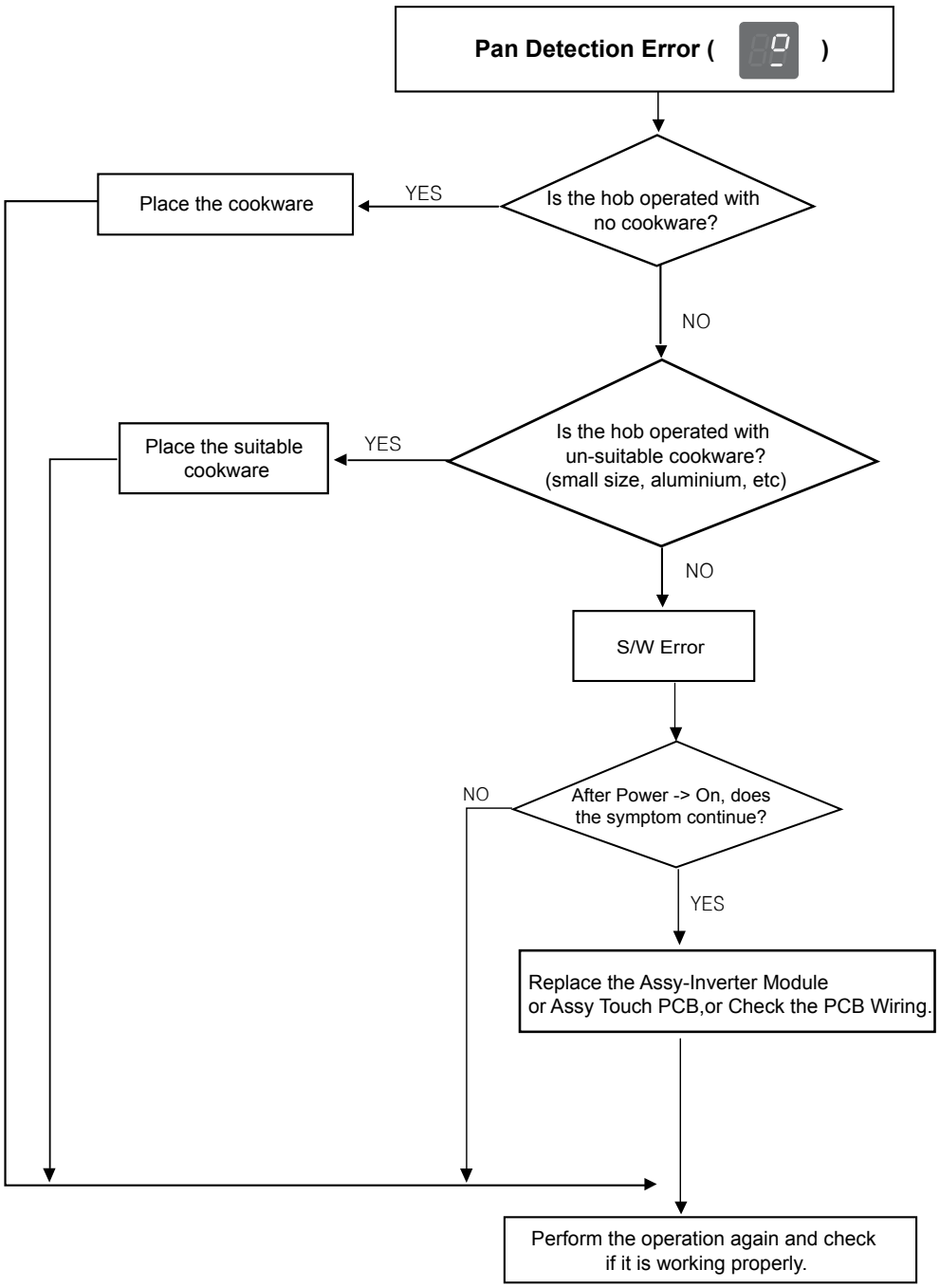
4-2-8 LE


Low Voltage Error (LE)
Is the sub wire connected?
(from Assy PCB Parts to Assy Touch PCB)
NO
Connect the sub wire.
YES
Is the sub wire sequence matched
with PCB schematic?
NO
Replace the sub wire.
YES
Is the SMPS's DC line has some
shorted circuit or opened circuit?
NO
S/W Error
YES
Assy PCB Parts has defect.
Replace the Assy PCB Parts .
After Power -> On, does
the symptom continue?
NO
YES
Replace the Assy PCB Parts.
Perform the operation again and check
if it is working properly.

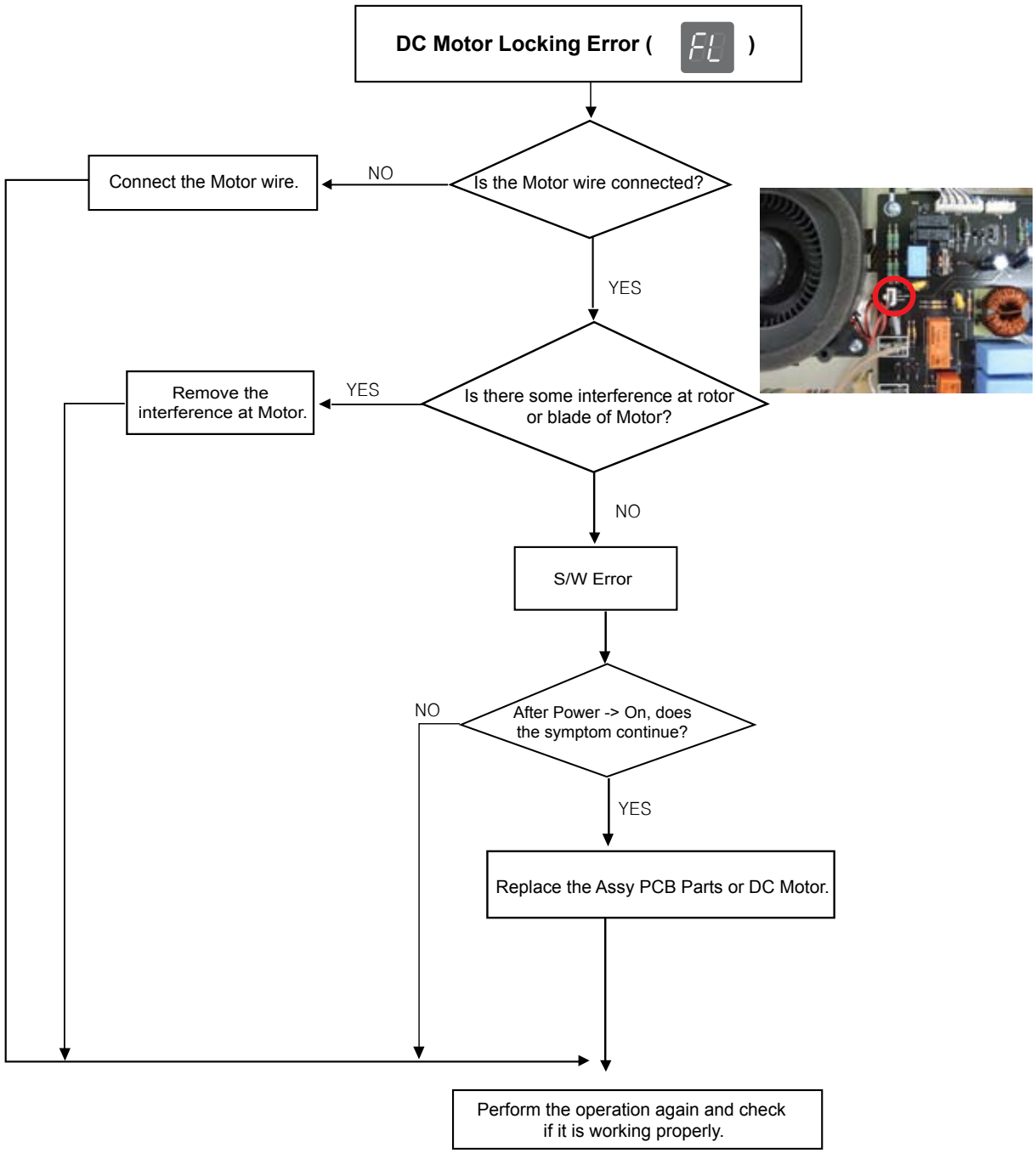
4-2-9 ot


Over Temperature Error (ot)
Is the hob operated with empty cookware?
YES
Make the hob be cool (Ceramic glass's surface)
NO
Is the Top sensor wire connected properly by burner position?
NO
Connect the wire properly.
YES
Is the Top sensor's real resistance is opened circuit?
NO
S/W Error
YES
The Top sensor has defect. Replace the sensor.
NO
After Power -> On, does the symptom continue?
YES
Replace the Assy Touch PCB.
Perform the operation again and check if it is working properly.

4-2-10 

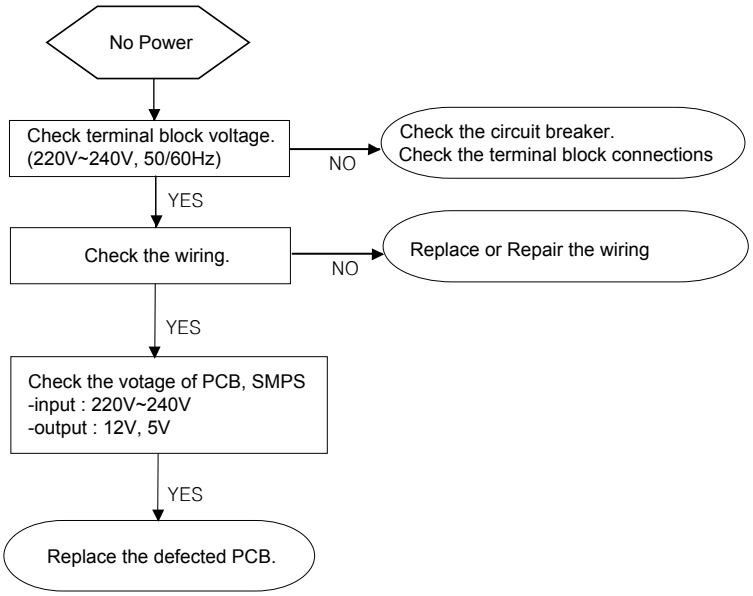

Pan Detection Error ( )
Is the hob operated with no cookware?
YES
Place the cookware
NO
Is the hob operated with un-suitable cookware? (small size, aluminium, etc)
YES
Place the suitable cookware
NO
S/W Error
After Power -> On, does the symptom continue?
NO
YES
Replace the Assy-Inverter Module or Assy Touch PCB,or Check the PCB Wiring.
Perform the operation again and check if it is working properly.

4-2-11 FL

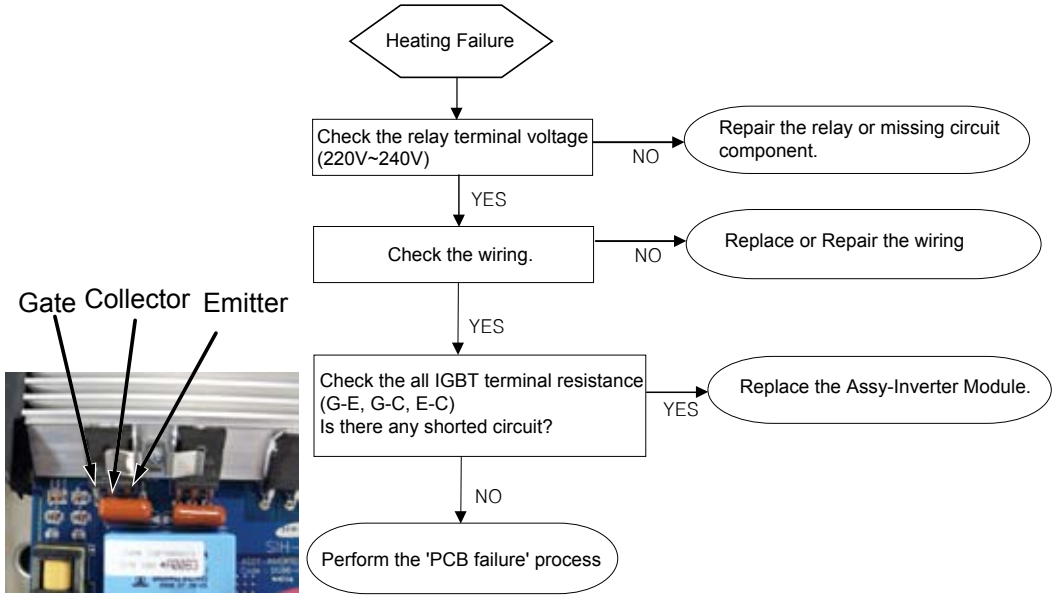

DC Motor Locking Error (FL)
Is the Motor wire connected?
NO
Connect the Motor wire.
YES
Is there some interference at rotor or blade of Motor?
YES
Remove the interference at Motor.
NO
S/W Error
After Power -> On, does the symptom continue?
NO
YES
Replace the Assy PCB Parts or DC Motor.
Perform the operation again and check if it is working properly.

4-3 Electrical Malfunction

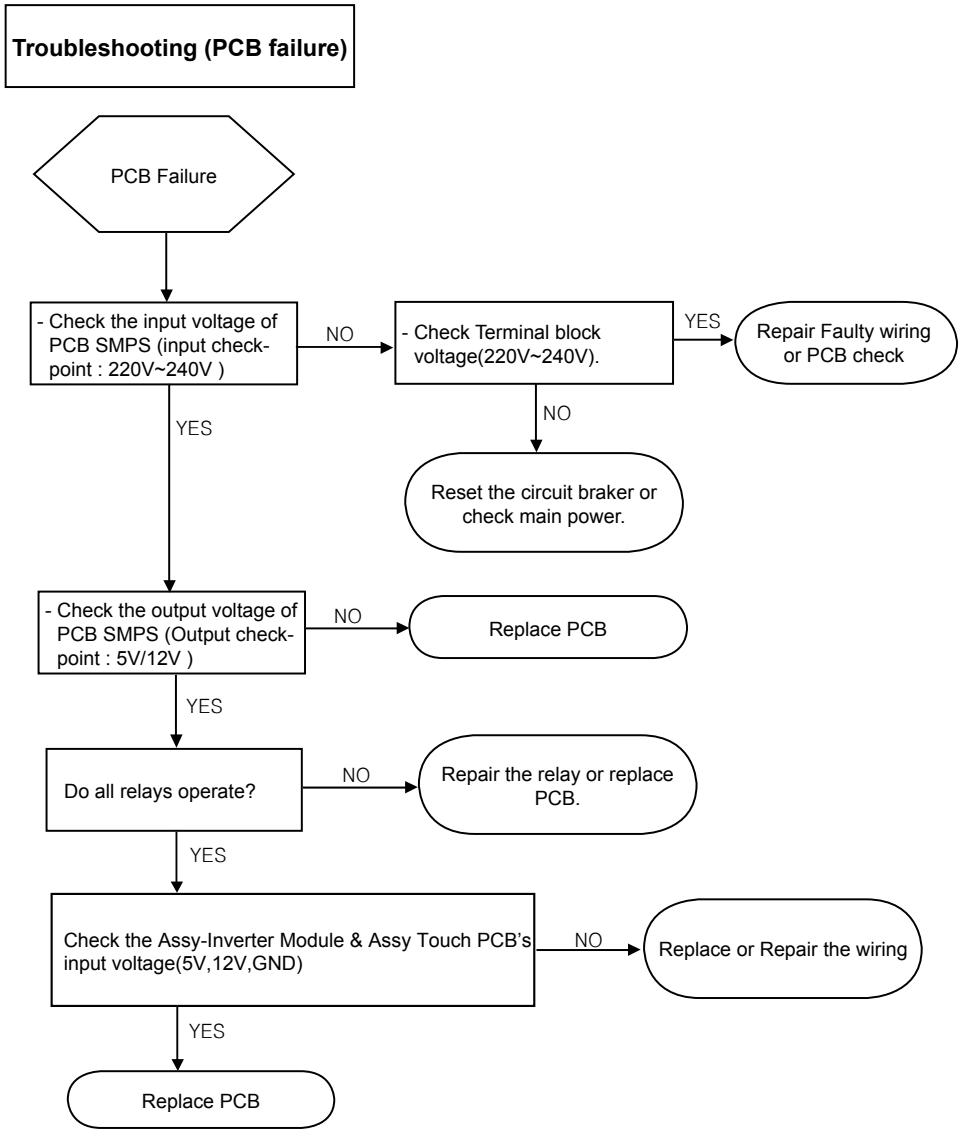
Troubleshooting (Power)



Troubleshooting (Heating)

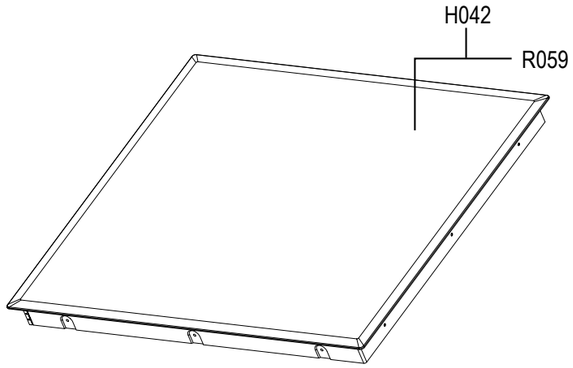


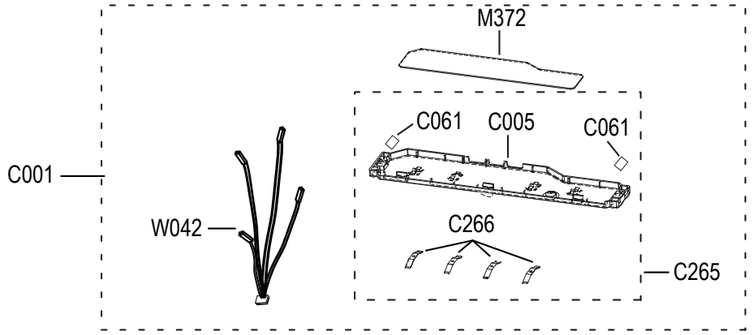
4-3 Electrical Malfunction

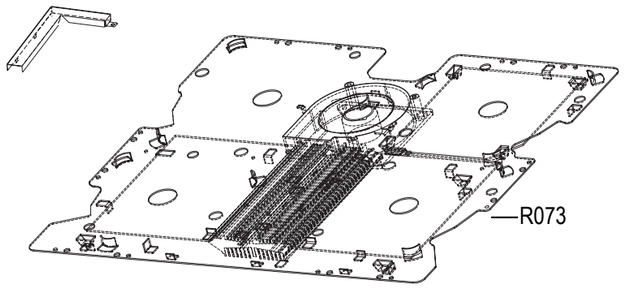


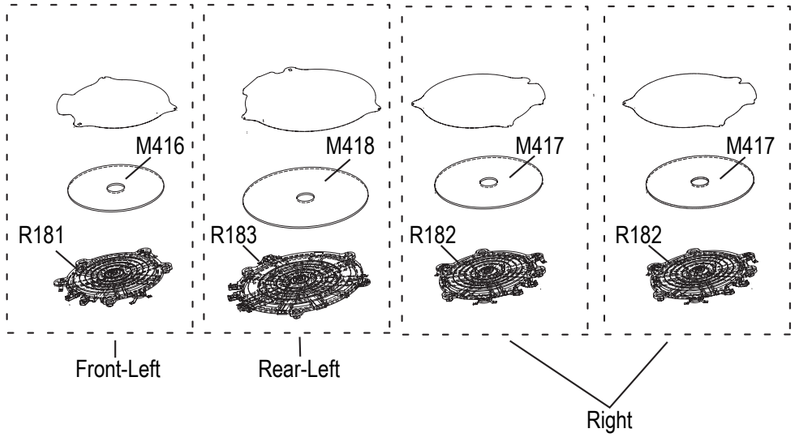
5. Exploded Views and Parts List

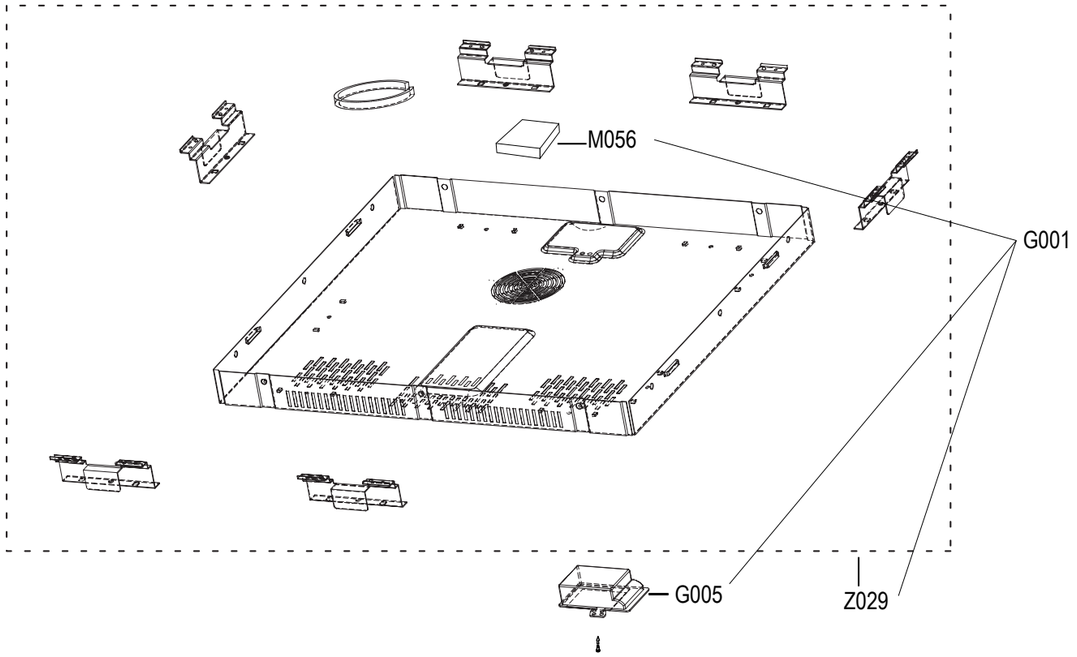
5-1 Exploded Views

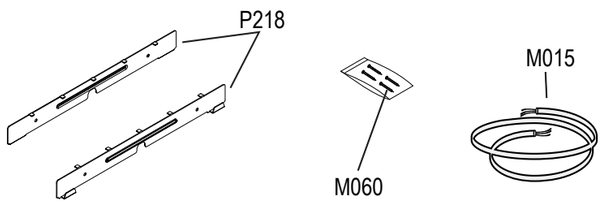












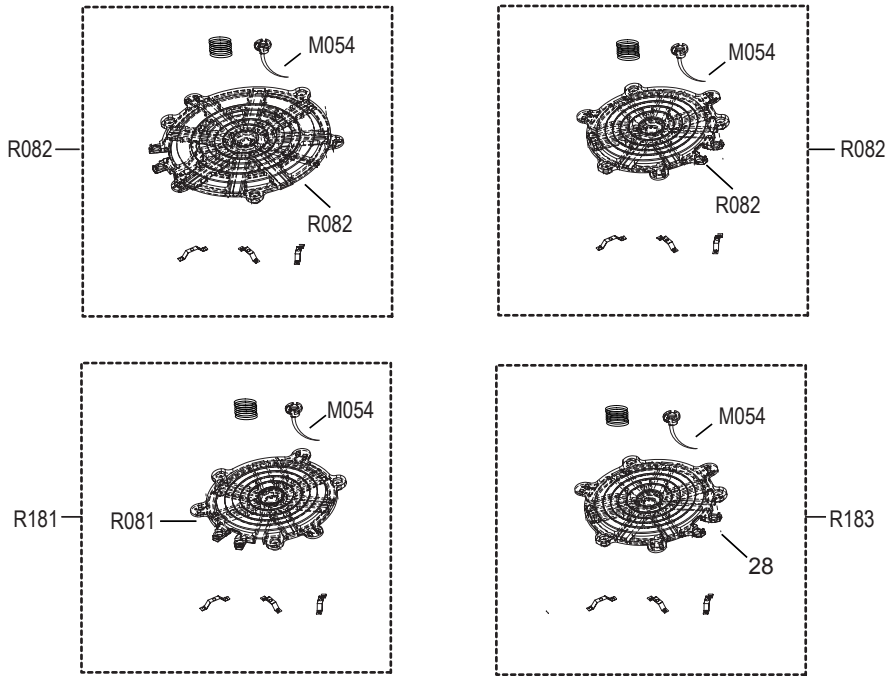
5-2 Main Parts List

(SNA : SERVICE NOT AVAILABLE)

No.	Code No.	Description	Specification	Q'ty	SA/ SNA	Remark
C001	DG94-00179A	ASSY CONTROL PANEL	CTI613EHST,-,-,BETTER	1	SA	-
C005	DG64-00213A	CONTROL-PANEL	CTI613EHST,PBT212G30V0,T2.	1	SNA	-
C061	0203-001815	TAPE-DOUBLE FACE	3M #4016,-,T1.6,W7,L14,	2	SNA	-
C265	DG94-00181A	ASSY-SPRING PANEL	CTI613EHST,-,INDUCTION	1	SA	-
C266	DG61-00274A	SPRING ETC-CONTROL	C61R2CCN,STS301,-,-,-	4	SNA	-
G001	DG94-00178A	ASSY CASE BURNER	CTI613EHST,CTI613GIN,IN	1	SA	-
G005	DG65-00006A	TERMINAL BLOCK	CTI613EHST,-,6P,450V,41A,	1	SNA	-
H042	DG94-00176A	ASSY TOP PLATE	CTI613EHST,BETTER DESIGN,	1	SA	-
M015	DG96-00061B	ASSY POWER CORD	VDE,Without Plug,250V/25	1	SA	-
M056	DE63-00005A	CUSHION-HVT	MW5896W,EPDM,T5,W24.7,L40,BL	1	SNA	-
M060	DG97-00010A	ASSY HARD WARE	C61RCCNR,-,NO-FRAME	1	SA	-
M372	DE96-00747B	ASSY TOUCH	CT1613,CT7000-0990,AMBER,310x	1	SA	-
M416	DG62-00028A	ADIABATIC-COIL A	CTI613EHST,S-F-INSULATI	1	SA	-
M417	DG62-00029A	ADIABATIC-COIL B	CTI613EHST,S-F-INSULATI	2	SA	-
M418	DG62-00030A	ADIABATIC-COIL C	CTI613EHST,S-F-INSULATI	1	SA	-
P218	DG61-00005B	BRACKET-INSTALL	C61RCCNR/BWT,ALSTAR,T1.0	2	SA	-
R069	DG64-00220A	GLASS-TOP PLATE	CTI613EHST,GLASS,T4.0,W5	1	SA	-
R073	DG96-00085A	ASSY-INDUCTION MODULE	CTI613EHST,-,-,-,-	1	SA	-
R181	DG96-00088A	ASSY-WORKING COIL A	CTI613EHST,-,-,-,-,1	1	SA	-
R182	DG96-00087A	ASSY-WORKING COIL B	CTI613EHST,-,-,-,-,1	2	SA	-
R183	DG96-00086A	ASSY-WORKING COIL C	CTI613EHST,-,-,-,-,2	1	SA	-
W042	DG96-00095A	ASSY WIRE HARNESS-S	CTI613EHST,230V50Hz,	1	SA	TOP SENSOR
Z029	DG94-00184A	ASSY CASE-SUPPORT	CTI613EHST,INDUCTION	1	SA	-

5-3 Assy-Working Coil Parts List

R181	ASSY-WORKING COIL A
R082	ASSY-WORKING COIL B
R083	ASSY-WORKING COIL C

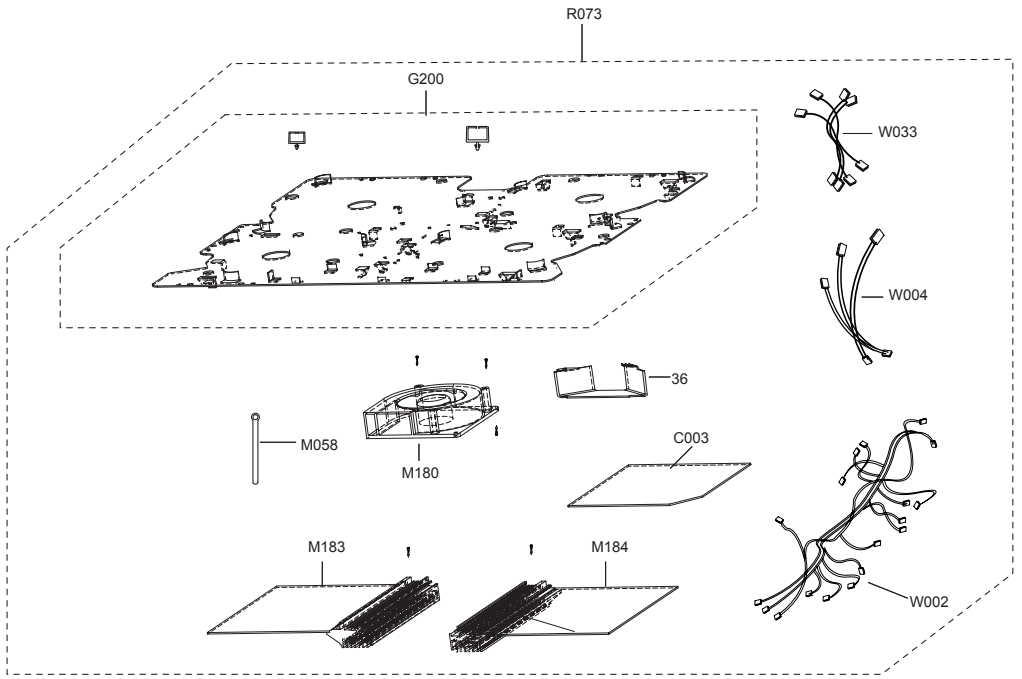


(SNA : SERVICE NOT AVAILABLE)

No.	Code No.	Description	Specification	Q'ty	SA/ SNA	Remark
M054	DG32-00005A	SENSOR-TOP	-,CTI613EHST,-,-,-,-,-,-,INDU	2	SA	-
M054	DG32-00005A	SENSOR-TOP	-,CTI613EHST,-,-,-,-,-,-,INDU	2	SA	-
R081	DG27-01003A	COIL-WORKING A	160pi, 1400W,97uH,#10%,Ba	1	SA	-
R082	DG27-01004A	COIL-WORKING B	180pi, 1800W,78uH,#10%,Ba	2	SA	-
R083	DG27-01005A	COIL-WORKING C	210pi, 2200W,62uH,#10%,Ba	1	SA	-
R181	DG96-00088A	ASSY-WORKING COIL A	CTI613EHST,-,-,-,-,1	1	SA	-
R182	DG96-00087A	ASSY-WORKING COIL B	CTI613EHST,-,-,-,-,1	2	SA	-
R183	DG96-00086A	ASSY-WORKING COIL C	CTI613EHST,-,-,-,-,2	1	SA	-

5-4 Assy-Induction Module Parts List

R073 ASSY-INDUCTION MODULE



(SNA : SERVICE NOT AVAILABLE)

No.	Code No.	Description	Specification	Q'ty	SA/ SNA	Remark
C003	RCS-ICTOP-01	ASSY PCB PARTS	CTI613EHST,230V50HZ,INDUC	1	SA	-
G200	DG94-00183A	ASSY CASE-INDUCTION	CTI613EHST,BETTER	1	SA	-
M058	6502-001085	CABLE CLAMP	DALC-4,ID5.3,T0.5,SGCC(GALVA	6	SNA	WORKING COIL WIRE
M180	DG31-00011A	MOTOR DC	D12F-12BM 06,CTI613EHST,0.83A,	1	SA	-
M183	DG96-00091A	ASSY-INVERTER MODULE L	CTI613EHST,-,-,-,	1	SA	-
M184	DG96-00092A	ASSY-INVERTER MODULE R	CTI613EHST,-,-,-,	1	SA	-
R073	DG96-00085A	ASSY-INDUCTION MODULE	CTI613EHST,-,-,-,-	1	SA	-
W002	DG96-00089A	ASSY WIRE HARNESS-A	CTI613EHST,230V50Hz,	1	SA	-
W004	DG96-00096A	ASSY WIRE HARNESS-C	CTI613EHST,230V50Hz,	1	SA	INVERTER, MAIN PCB
W033	DG96-00090A	ASSY WIRE HARNESS-P	CTI613EHST,230V50Hz,	1	SA	TERMINAL BLOCK

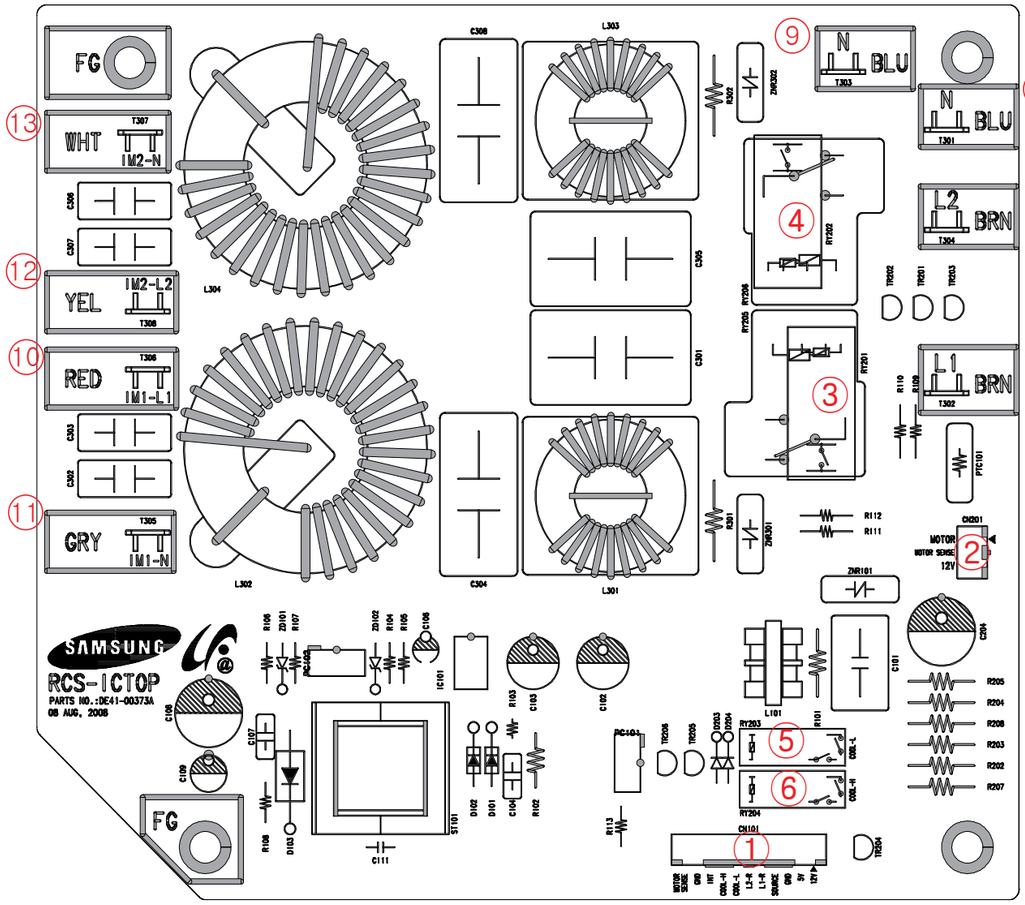
5-4 Standard Parts List

(SNA : SERVICE NOT AVAILABLE)

LEVEL	Code No.	Description	Specification	Q'ty	SA/ SNA	Remark
1-1	6002-000217	SCREW-TAPPING	TH,+,-,1,M4,L8,ZPC(WHT),SW	25	SNA	-
1-2	6006-001170	SCREW-TAPPING	TH,+,WT,TC,M4,L10,ZPC(WHT)	1	SNA	EARTH
1-3	6002-000217	SCREW-TAPPING	TH,+,-,1,M4,L8,ZPC(WHT),SW	6	SNA	SUPPORT-HEATER
1-2	6002-000217	SCREW-TAPPING	TH,+,-,1,M4,L8,ZPC(WHT),SW	9	SNA	PCB,GUIDE-AIR
1-2	6002-000630	SCREW-TAPPING	PH,+,2S,M3,L8,ZPC(YEL),SWR	2	SNA	HEAT SINK
1-2	6002-001321	SCREW-TAPPING	PWH,+,2S,M5,L10,ZPC(YEL)	2	SNA	DC MOTOR
1-2	6006-001170	SCREW-TAPPING	TH,+,WT,TC,M4,L10,ZPC(WHT)	1	SNA	DC MOTOR
1-2	6002-000216	SCREW-TAPPING	TH,+,-,1,M4.0,L20,ZPC(WHT)	4	SNA	Bkt-install

6. P.C.B Diagrams

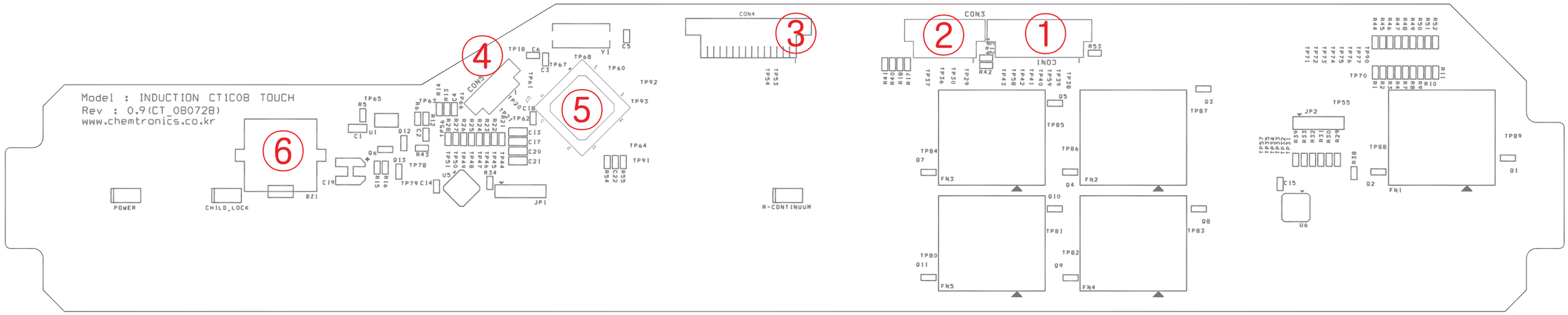
6-1 P.C.B Diagrams : ASSY PCB PARTS (This Document can not be used without Samsung's authorization)



No.	Parts Number	Part Name	Function and Rule
1	CN101	Main PCB wire connector	communication wire connector (to Assy Touch PCB)
2	CN201	DC Motor connector	Voltage input of DC Motor
3	RY201	Main Relay	Relay for L1-line input line
4	RY202	Main Relay	Relay for L2-line input line
5	RY203	DC Motor Relay (Low)	Relay for DC Motor control (low fan)
6	RY204	DC Motor Relay (High)	Relay for DC Motor control (high fan)
7	T302	Terminal	L1-line input from Terminal Block
8	T304	Terminal	L2-line input from Terminal Block
9	T301,303	Terminal	N-line input from Terminal Block
10	T306	Terminal	L1-line to Assy-Inverter Module R
11	T305	Terminal	N-line to Assy-Inverter Module R
12	T308	Terminal	L2-line to Assy-Inverter Module L
13	T307	Terminal	N-line to Assy-Inverter Module L

6. P.C.B Diagrams

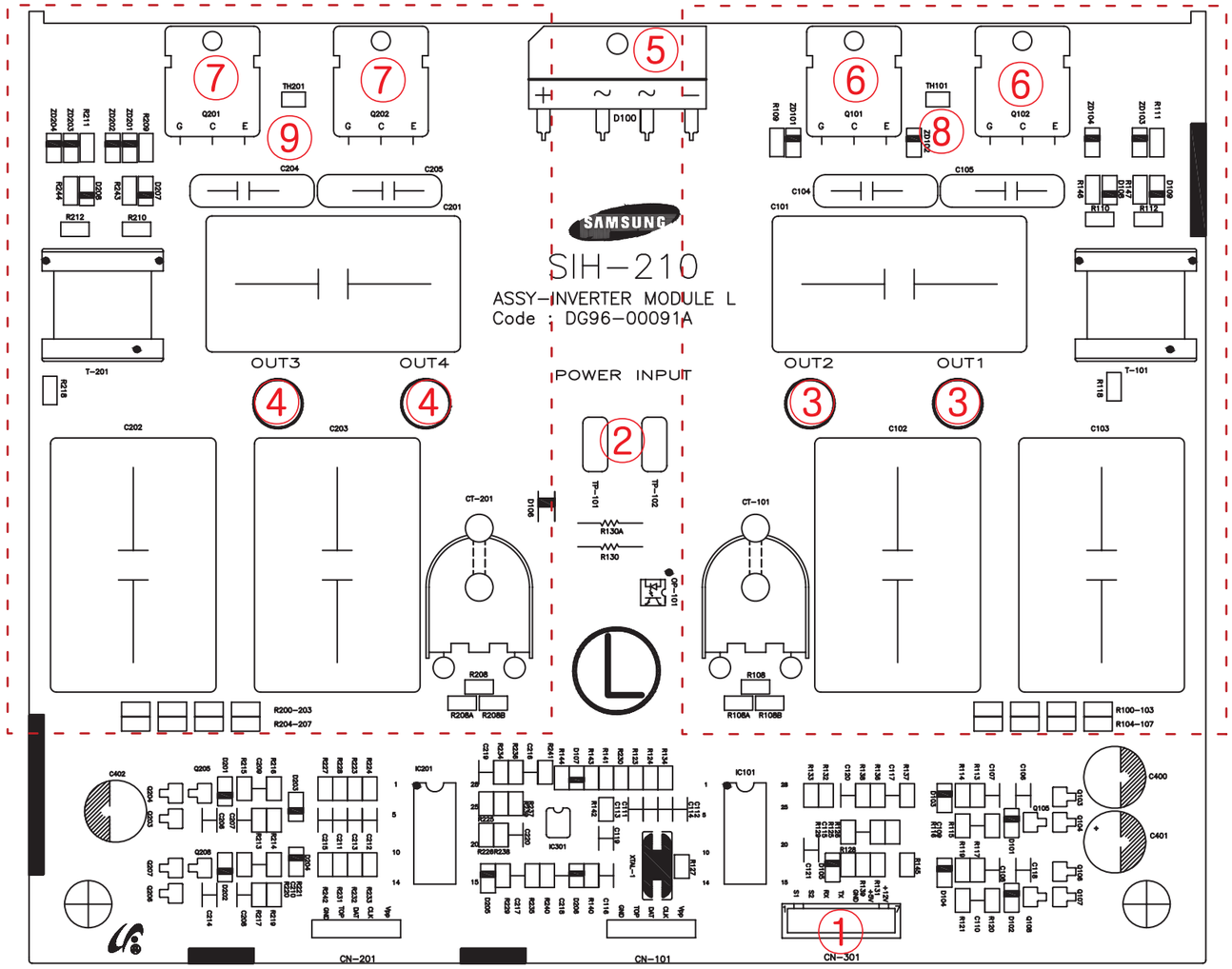
6-2 P.C.B Diagrams : ASSY TOUCH PCB (This Document can not be used without Samsung's authorization)



No.	Parts Number	Part Name	Function and Rule
1	CON1	Main PCB wire connector	communication wire connector (to Assy PCB Parts)
2	CON3	Top Sensor wire connector	sensor wire connector (to Sensor-Top)
3	CON4	Inverter PCB connector	communication wire connector (to Assy-Inverter Module)
4	CON5	MICOM Program connector	MICOM program input port, not connected
5	U7	MICOM	Main MICOM
6	BZ1	Buzzer	Beep Sound

6. P.C.B Diagrams

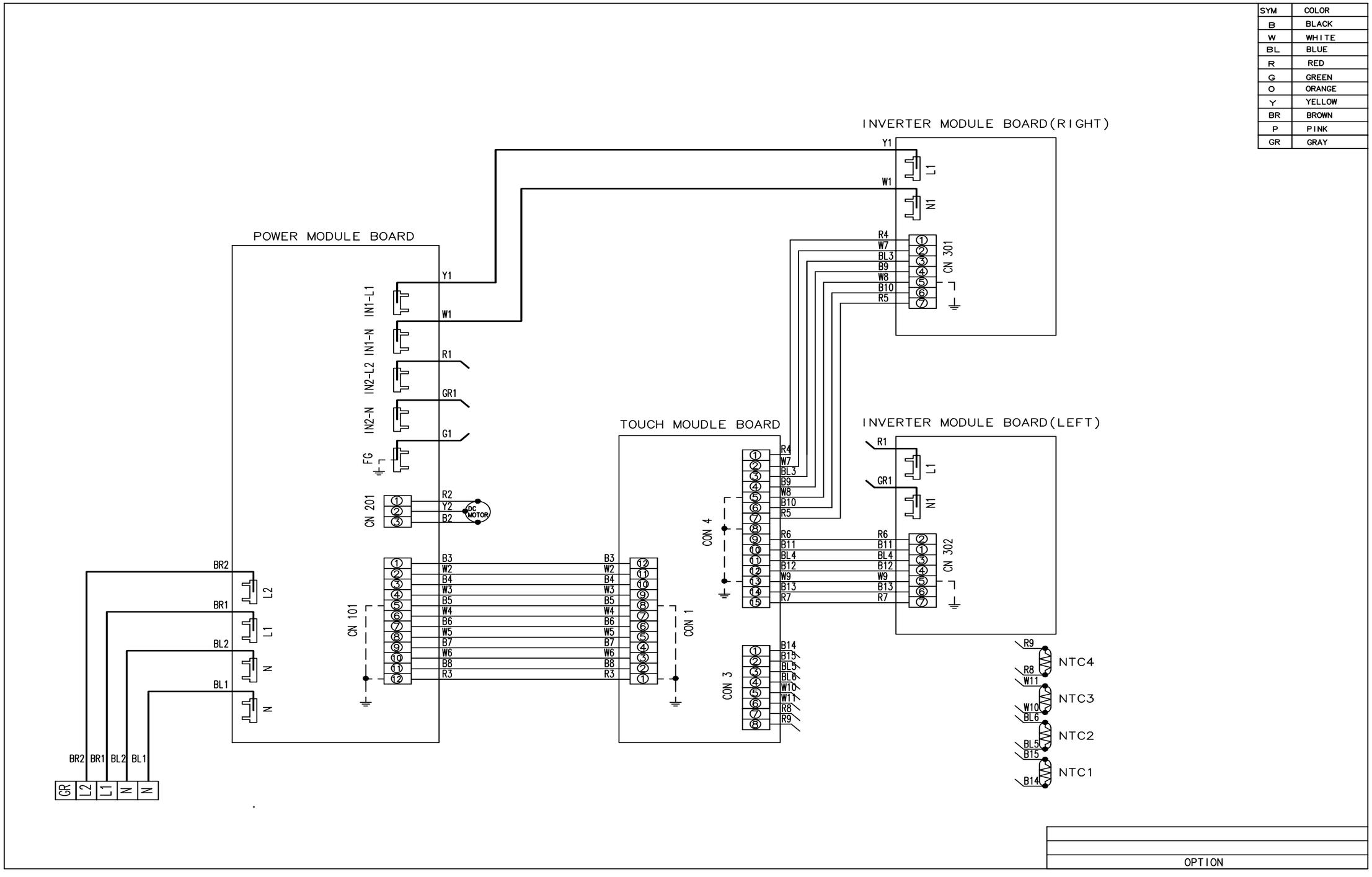
6-3 P.C.B Diagrams : ASSY-INVERTER MODULE PCB (This Document can not be used without Samsung's authorization)



No.	Parts Number	Part Name	Function and Rule
1	CN301	Wire connector	communication with Touch PCB, DC 5V,12V input port
2	TP-101,102	AC line terminal	AC voltage input terminal
3	OUT1,2	Working-Coil wire	High frequency AC voltage output wire (right circuit)
4	OUT3,4	Working-Coil wire	High frequency AC voltage output wire (left circuit)
5	D100	Bridge IC	Full-bridge rectifier IC
6	Q101,102	IGBT IC	High frequency switching IC (right circuit)
7	Q201,202	IGBT IC	High frequency switching IC (left circuit)
8	TH101	IGBT NTC sensor	Thermistor for IGBT temp (right circuit)
9	TH201	IGBT NTC sensor	Thermistor for IGBT temp (left circuit)

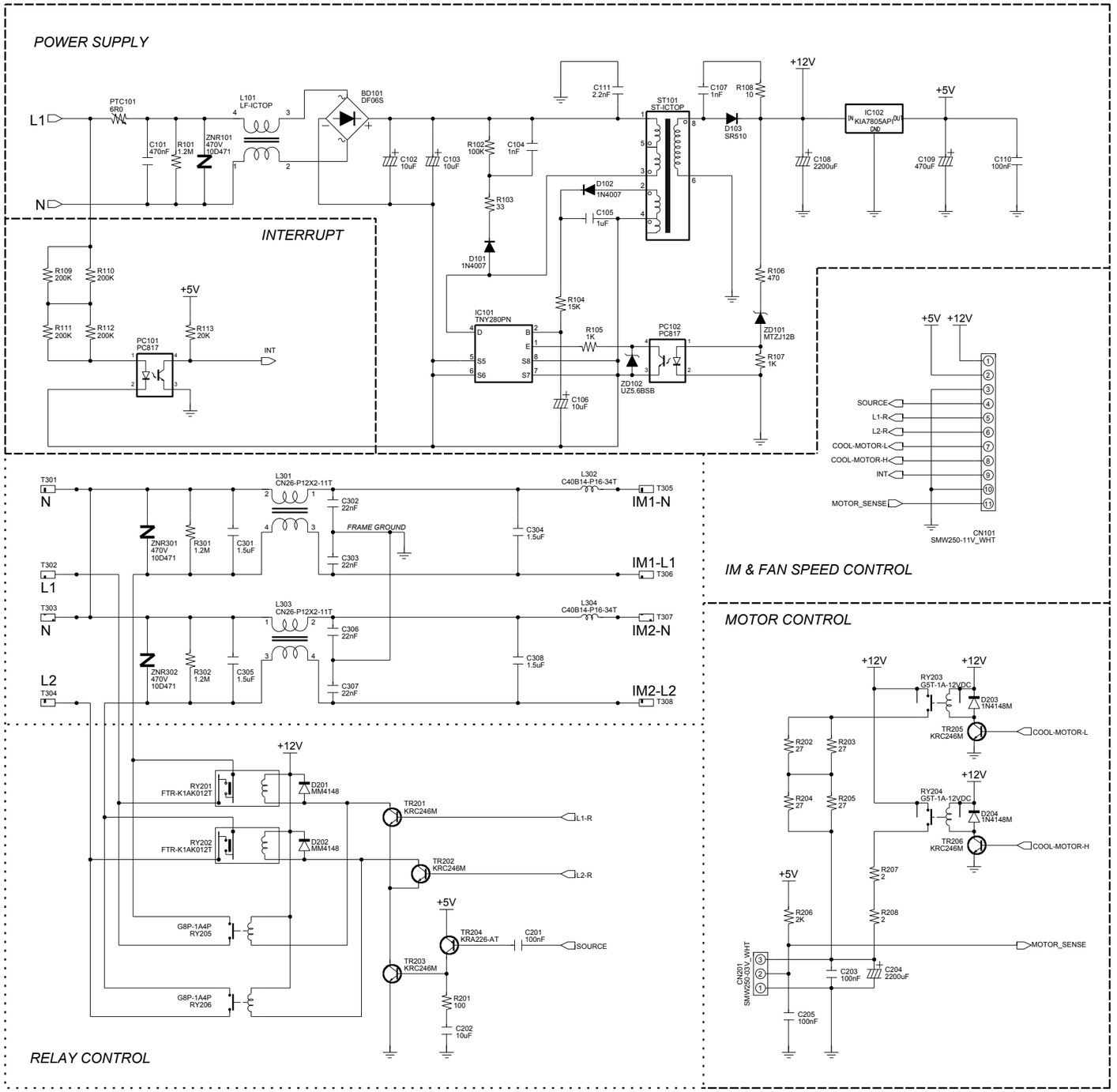
7. Wiring Diagrams (This Document can not be used without Samsung's authorization)

7-1 Wiring Diagrams



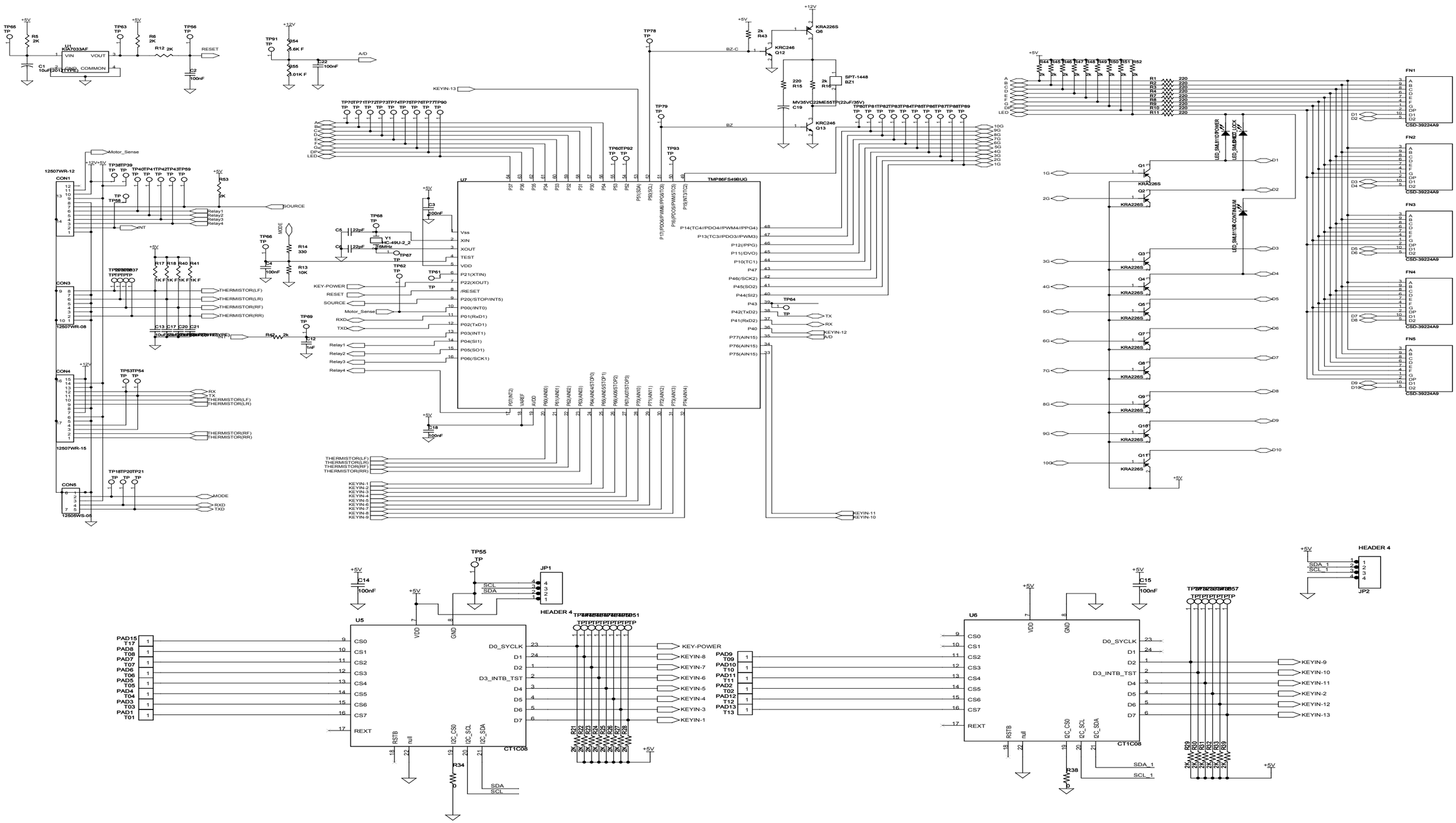
8. Schematic Diagrams

8-1 Schematic Diagrams : ASSY PCB PARTS (This Document can not be used without Samsung's authorization)



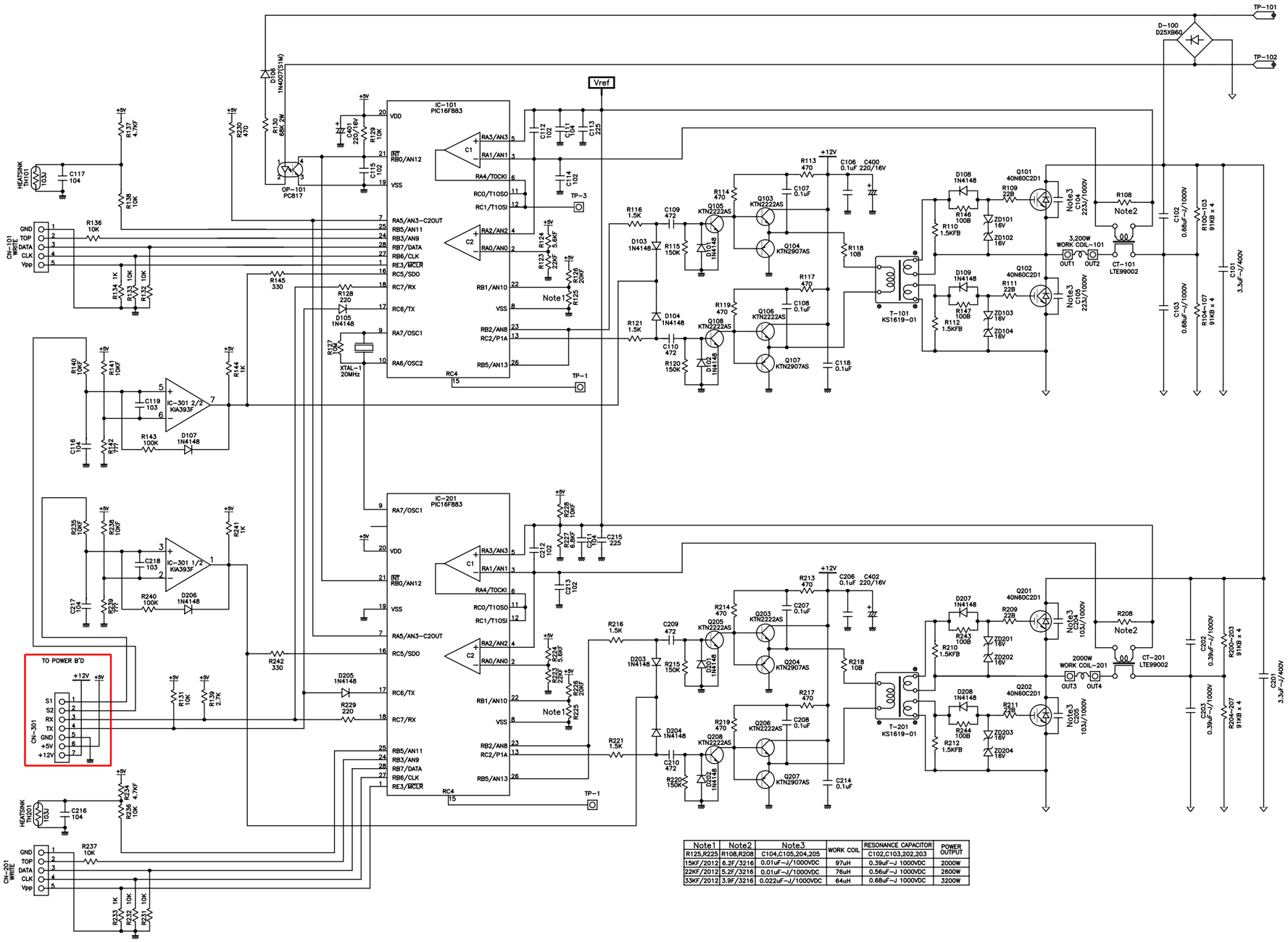
8. Schematic Diagrams

8-2 Schematic Diagrams : ASSY TOUCH PCB (This Document can not be used without Samsung's authorization)



8. Schematic Diagrams

8-3 Schematic Diagrams : ASSY-INVERTER MODULE PCB (This Document can not be used without Samsung's authorization)



Note1	Note2	Note3	WORK COIL	RESONANCE CAPACITOR	POWER OUTPUT
R125,R225	R108,R208	C104,C105,204,205		C102,C103,202,203	
15KF/2012	6.2F/3216	0.01uF~J/1000VDC	97uH	0.39uF~J 1000VDC	2000W
22KF/2012	5.2F/3216	0.01uF~J/1000VDC	76uH	0.56uF~J 1000VDC	2600W
33KF/2012	3.9F/3216	0.022uF~J/1000VDC	64uH	0.68uF~J 1000VDC	3200W

SAMSUNG ELECTRONICS

GSPN (Global Service Partner Network)

Contry	Web Site
North America	service.samsungportal.com
Latin America	latin.samsungportal.com
CIS	cis.samsungportal.com
Europe	europe.samsungportal.com
China	china.samsungportal.com
Asia	asia.samsungportal.com
Mideast & Africa	mea.samsungportal.com

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